

The Frequency Domain, without tears



Many
slides
borrowed
from
Steve
Seitz

Somewhere in Cinque Terre, May 2005

CS180: Intro to Computer Vision and Comp. Photo
Alexei Efros and Ren Ng, UC Berkeley, Fall 2026

Recap: Image filtering

$$h[\cdot, \cdot] \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

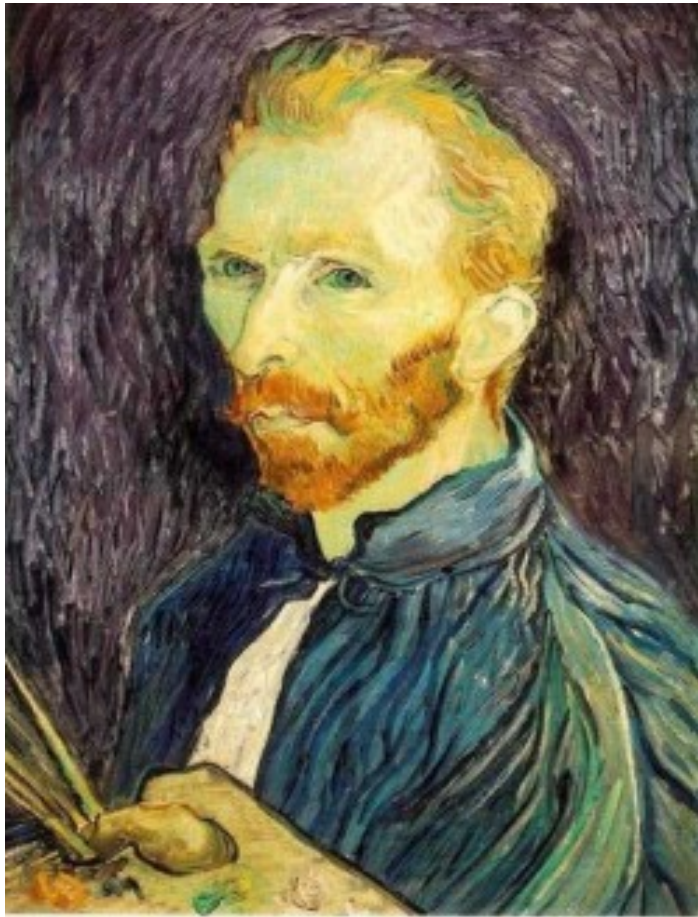
$f[\cdot, \cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$g[\cdot, \cdot]$

	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	30	60	90	90	90	60	30	
	0	30	50	80	80	90	60	30	
	0	30	50	80	80	90	60	30	
	0	20	30	50	50	60	40	20	
	10	20	30	30	30	30	20	10	
	10	10	10	0	0	0	0	0	

Filtering for antialiasing (low pass)



Gaussian $1/2$



G $1/4$



G $1/16$

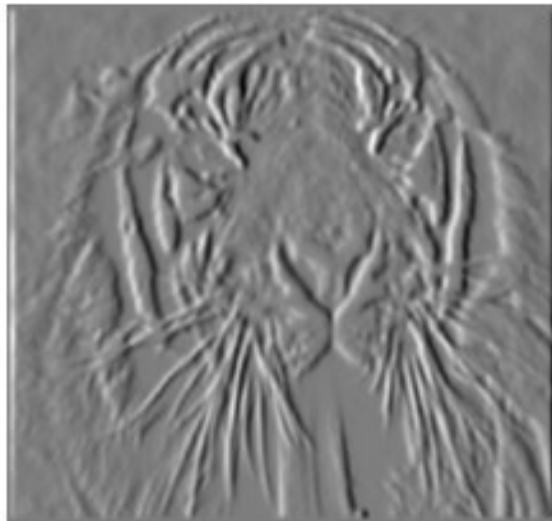
Solution: filter the image, *then* subsample

- Filter size should double for each $1/2$ size reduction. Why?

Filtering to compute image derivatives

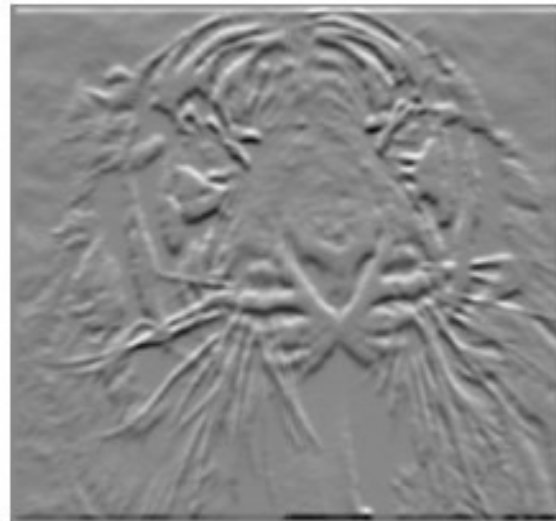


$$\frac{\partial f(x, y)}{\partial x}$$



-1	1
----	---

$$\frac{\partial f(x, y)}{\partial y}$$



-1
1

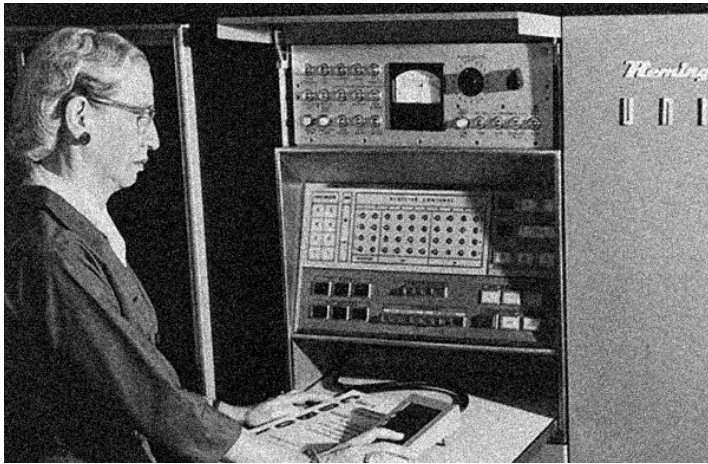
 or

1
-1

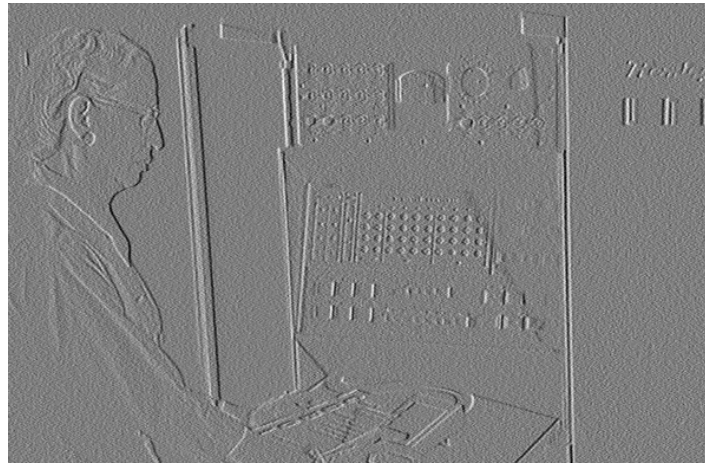
Which shows changes with respect to x?

Filtering to denoise

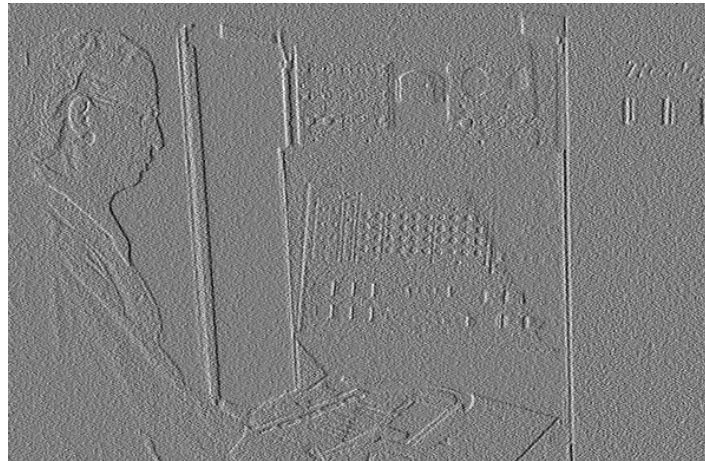
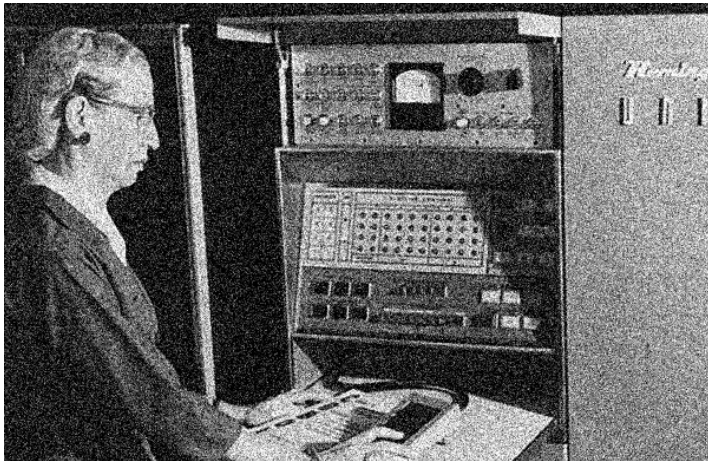
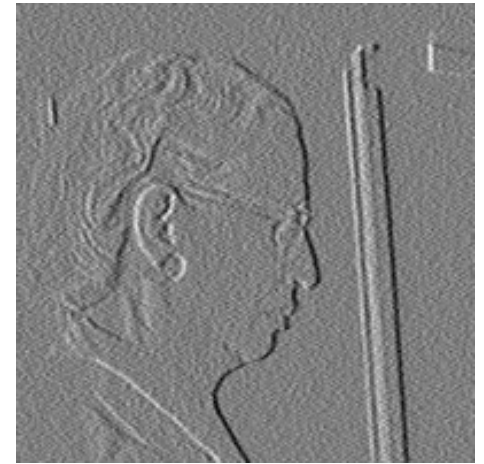
Noisy Input



dx via $[-1,0,1]$



Zoom



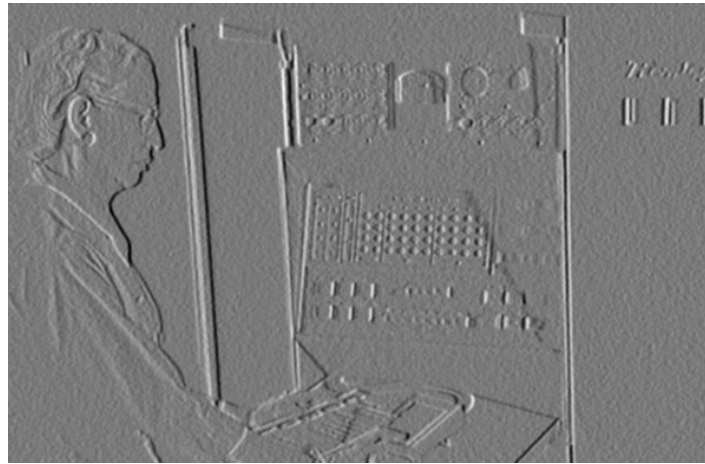
Source: D. Fouhey

Filtering to denoise

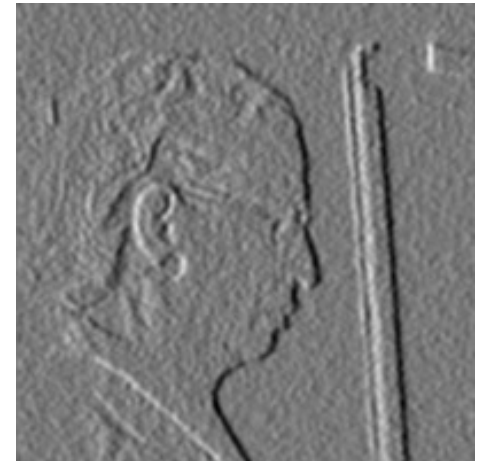
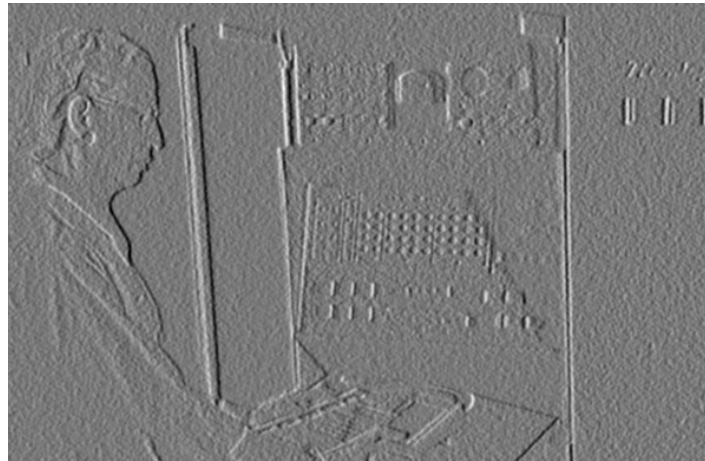
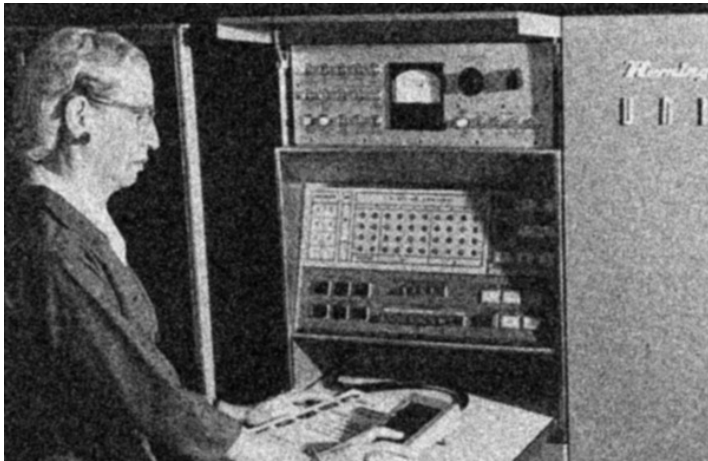
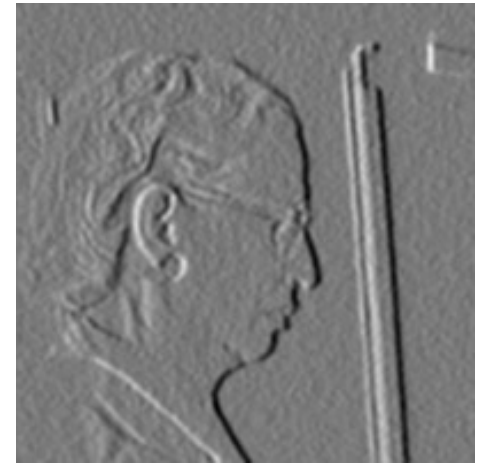
Smoothed Input



dx via $[-1,0,1]$

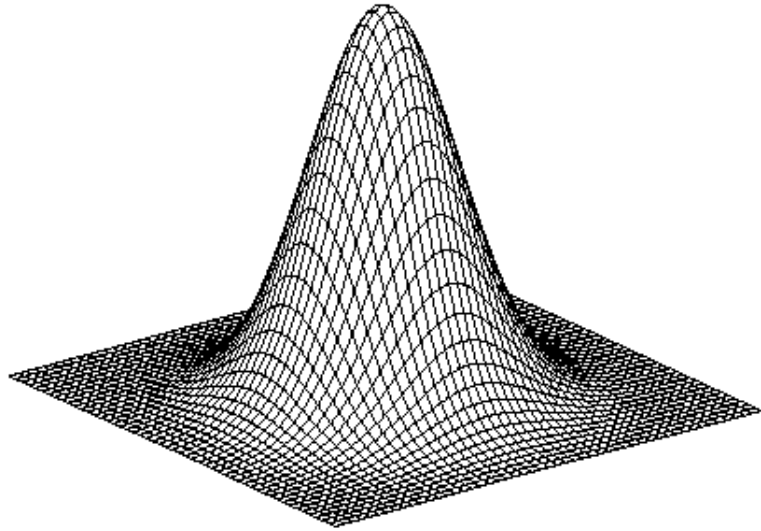


Zoom

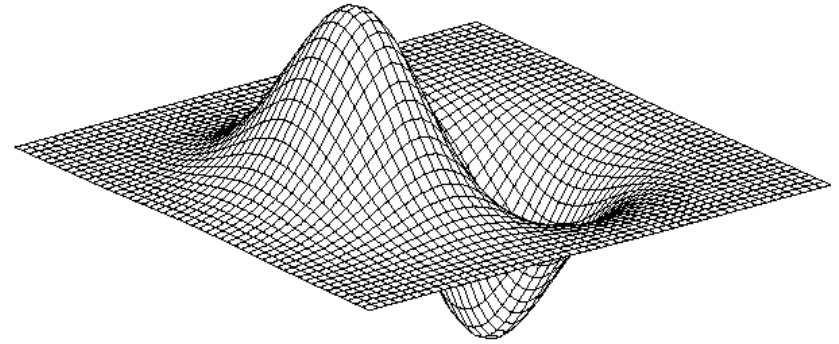


Source: D. Fouhey

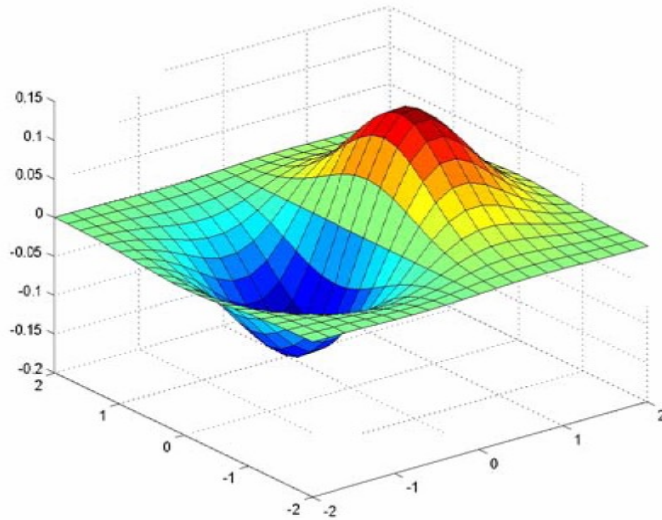
Derivative of Gaussian filter



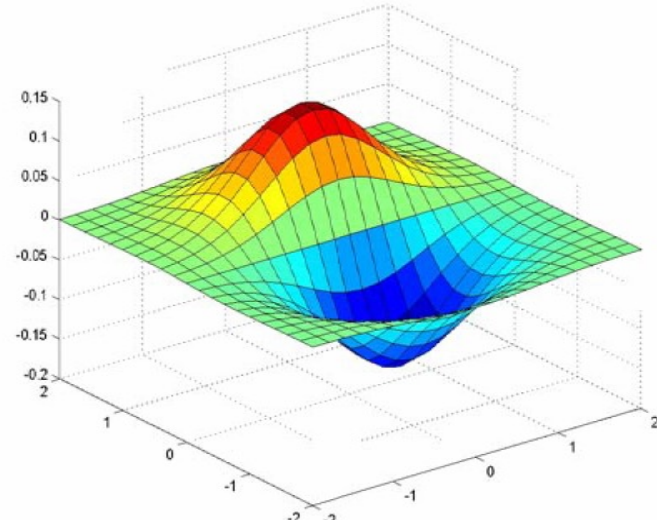
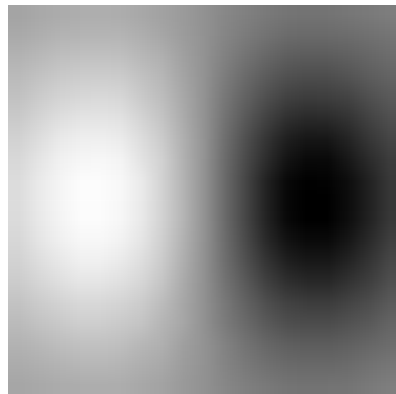
$$* [-1 \ 0 \ 1] =$$



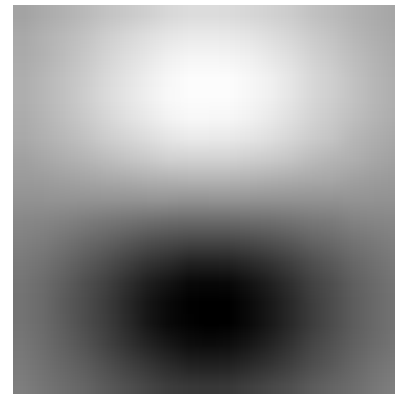
Derivative of Gaussian filter



x-direction



y-direction



Which one finds horizontal/vertical edges?

Low Pass vs. High Pass filtering

Image



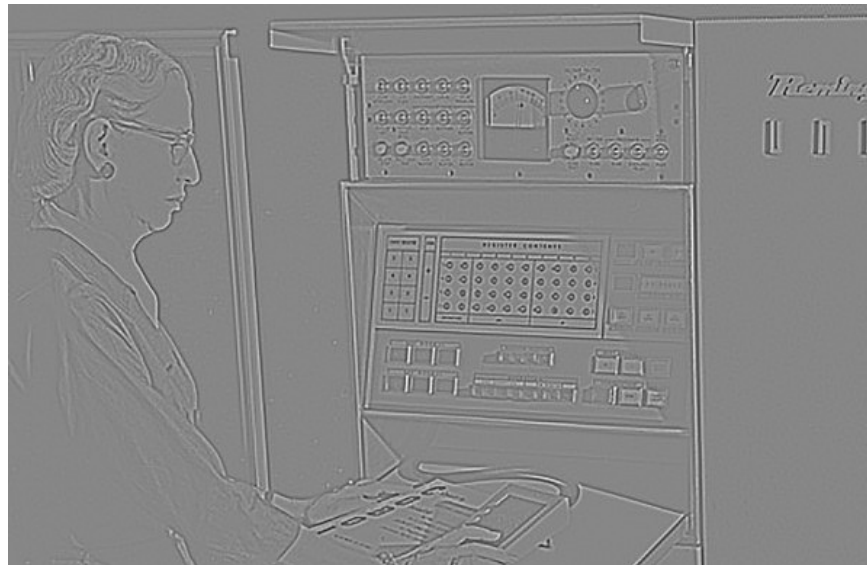
Smoothed



—

Details

=



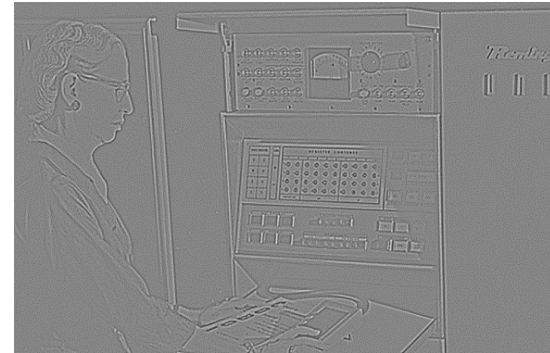
Filtering – Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=1$

$=$



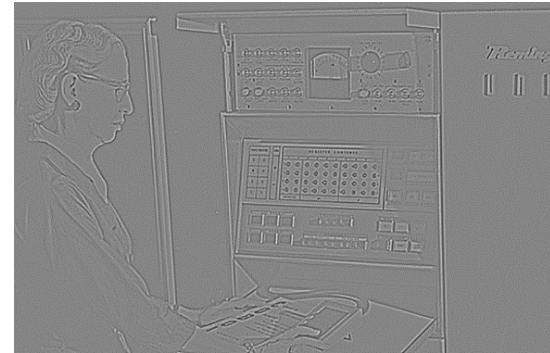
Filtering – Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=0$

$=$



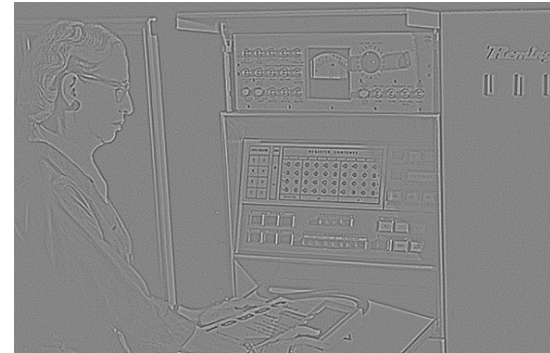
Filtering – Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=2$

$=$



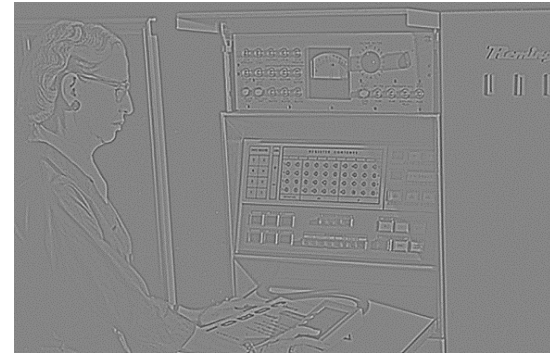
Filtering – Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=0$

$=$



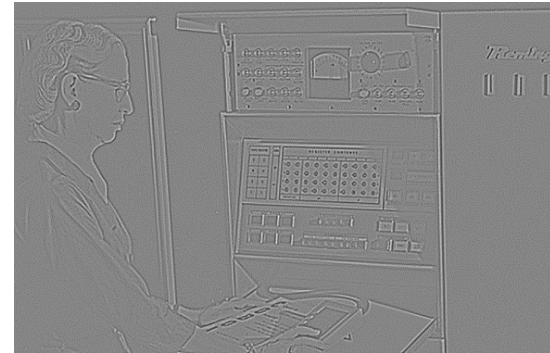
Filtering – Extreme Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=10$

=



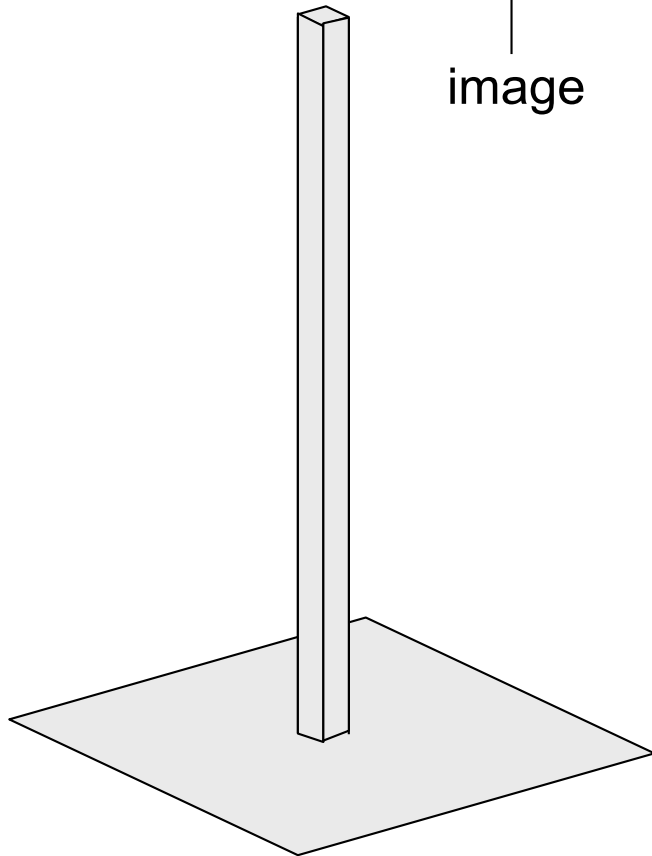
Unsharp mask filter (= sharpening filter)

$$f + \alpha(f - f * g) = (1 + \alpha)f - \alpha f * g = f * ((1 + \alpha)e - \alpha g)$$

↑
image

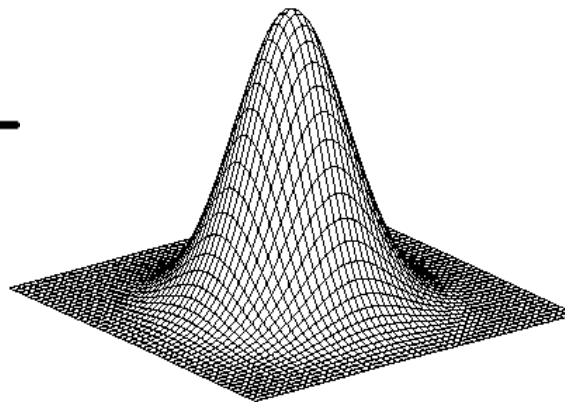
↑
blurred
image

↑
unit impulse
(identity)



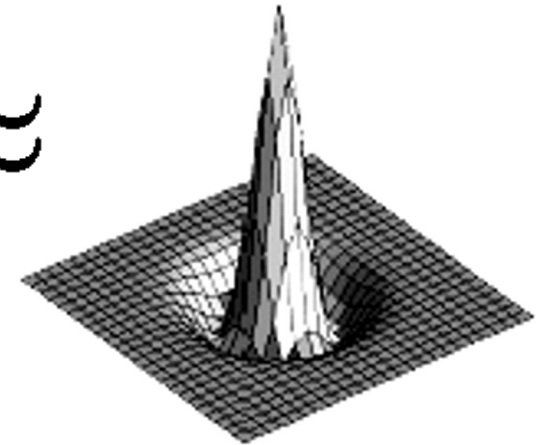
unit impulse

—



Gaussian

≈



Laplacian of Gaussian

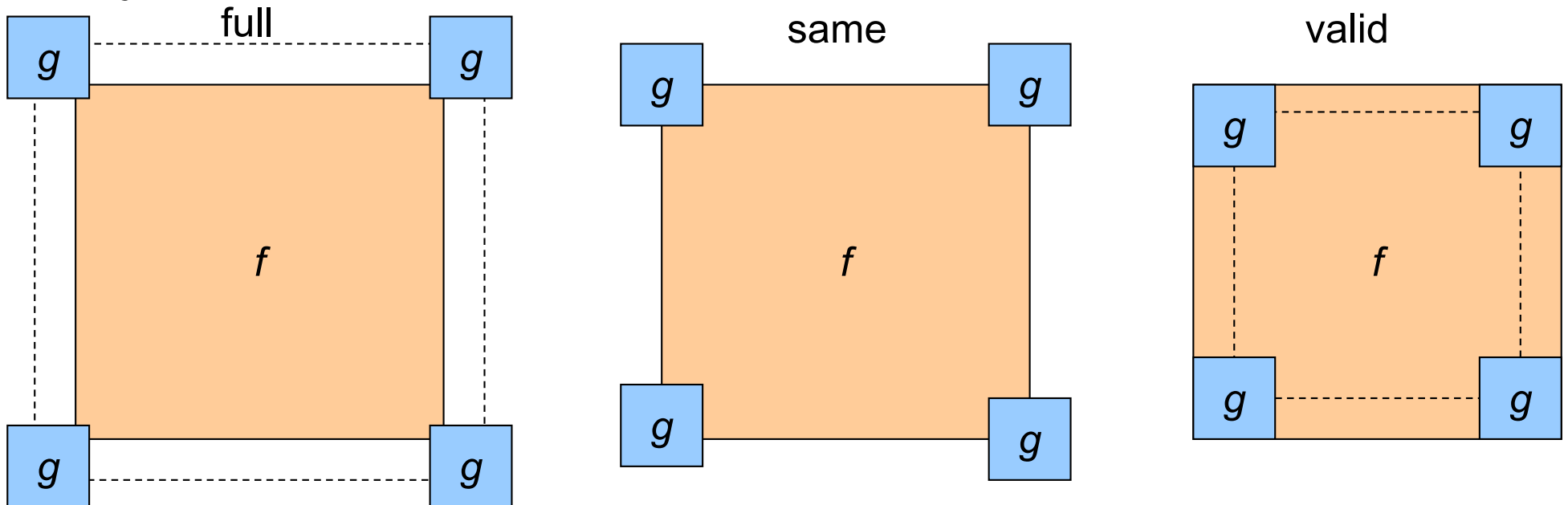
Filtering: practical matters

What is the size of the output?

(MATLAB) `filter2(g, f, shape)` or `conv2(g,f,shape)`

- *shape* = 'full': output size is sum of sizes of *f* and *g*
- *shape* = 'same': output size is same as *f*
- *shape* = 'valid': output size is difference of sizes of *f* and *g*

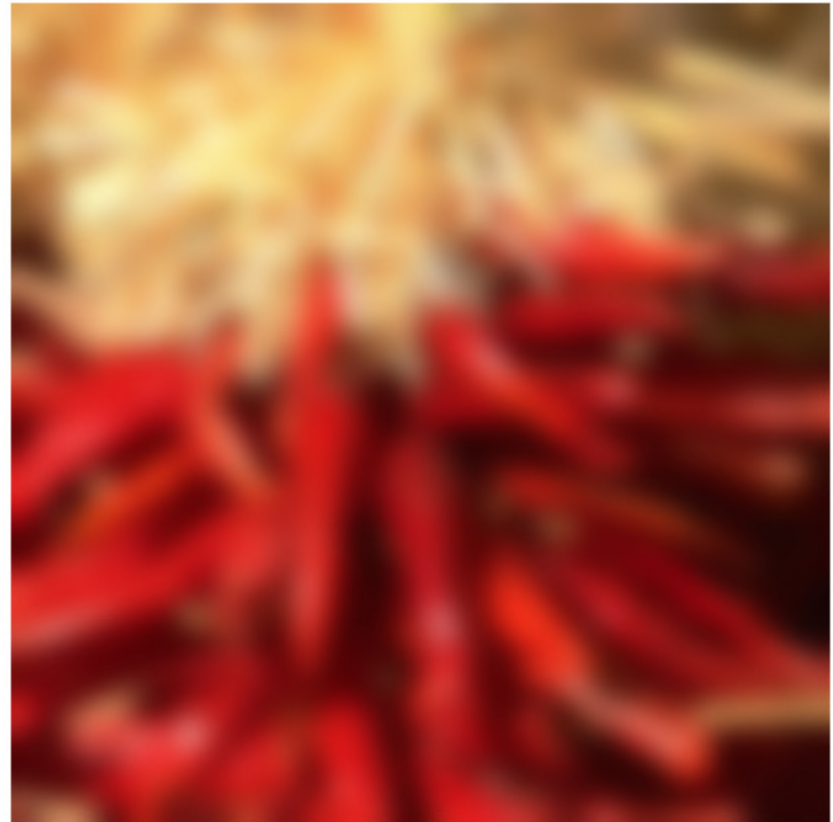
Pytorch `conv2d` 'valid' or 'same'



Filtering: practical matters

What about near the edge?

- the filter window falls off the edge of the image
- need to extrapolate
- methods:
 - clip filter (black)
 - wrap around (circular)
 - copy edge
 - reflect across edge



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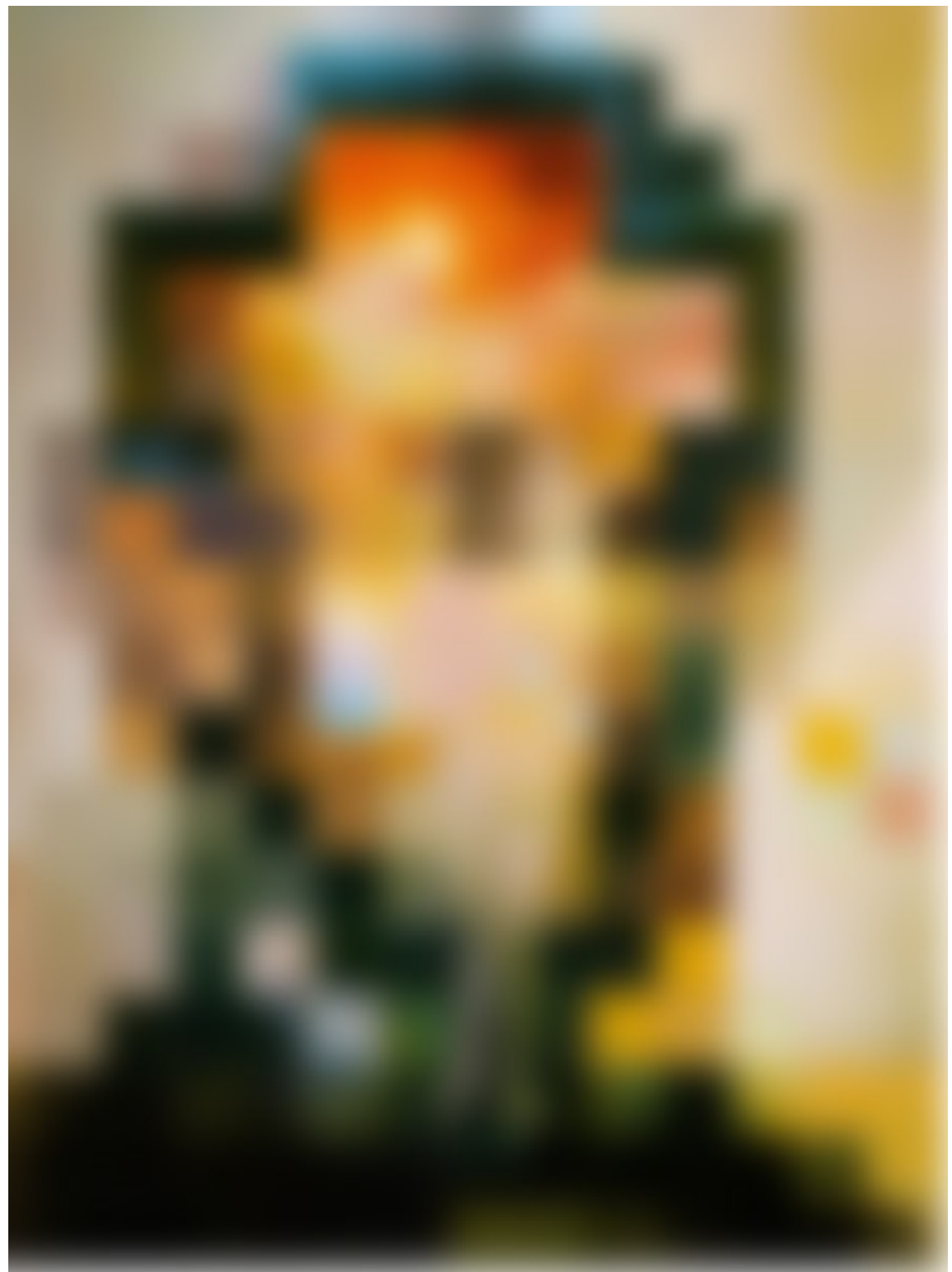
Somewhere in Cinque Terre, May 2005

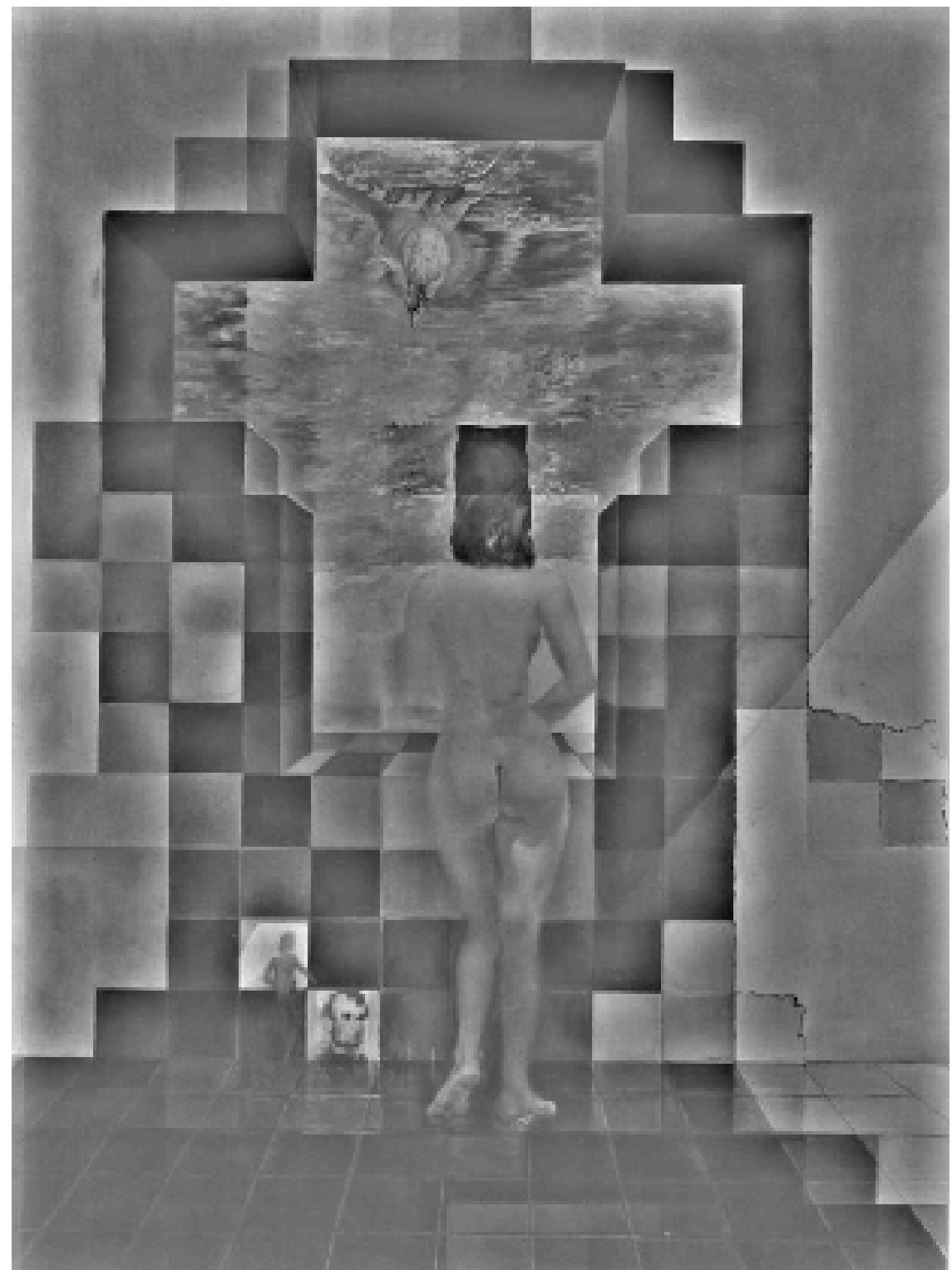
CS180: Intro to Computer Vision and Comp. Photo
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Salvador Dali

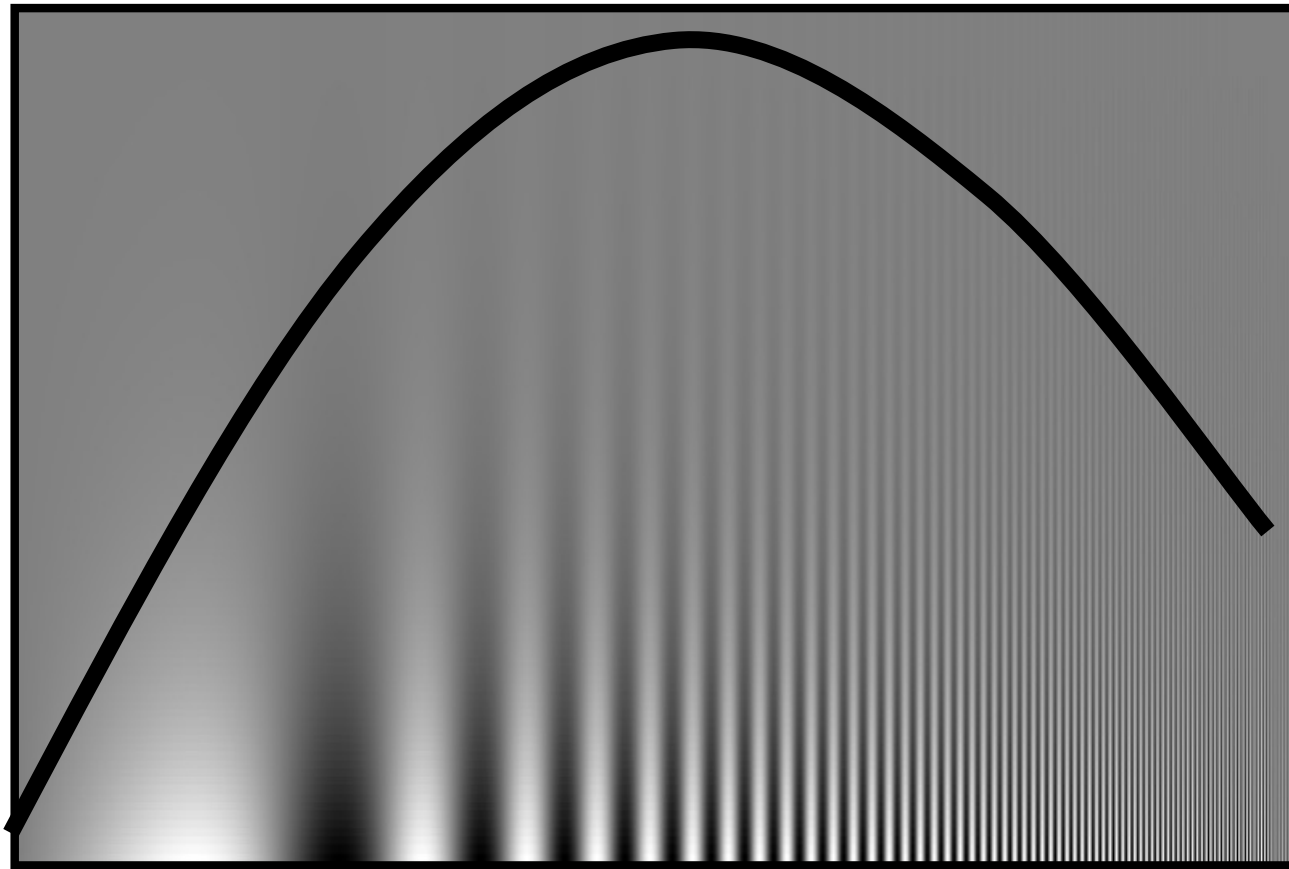
*"Gala Contemplating the Mediterranean Sea,
which at 30 meters becomes the portrait
of Abraham Lincoln", 1976*







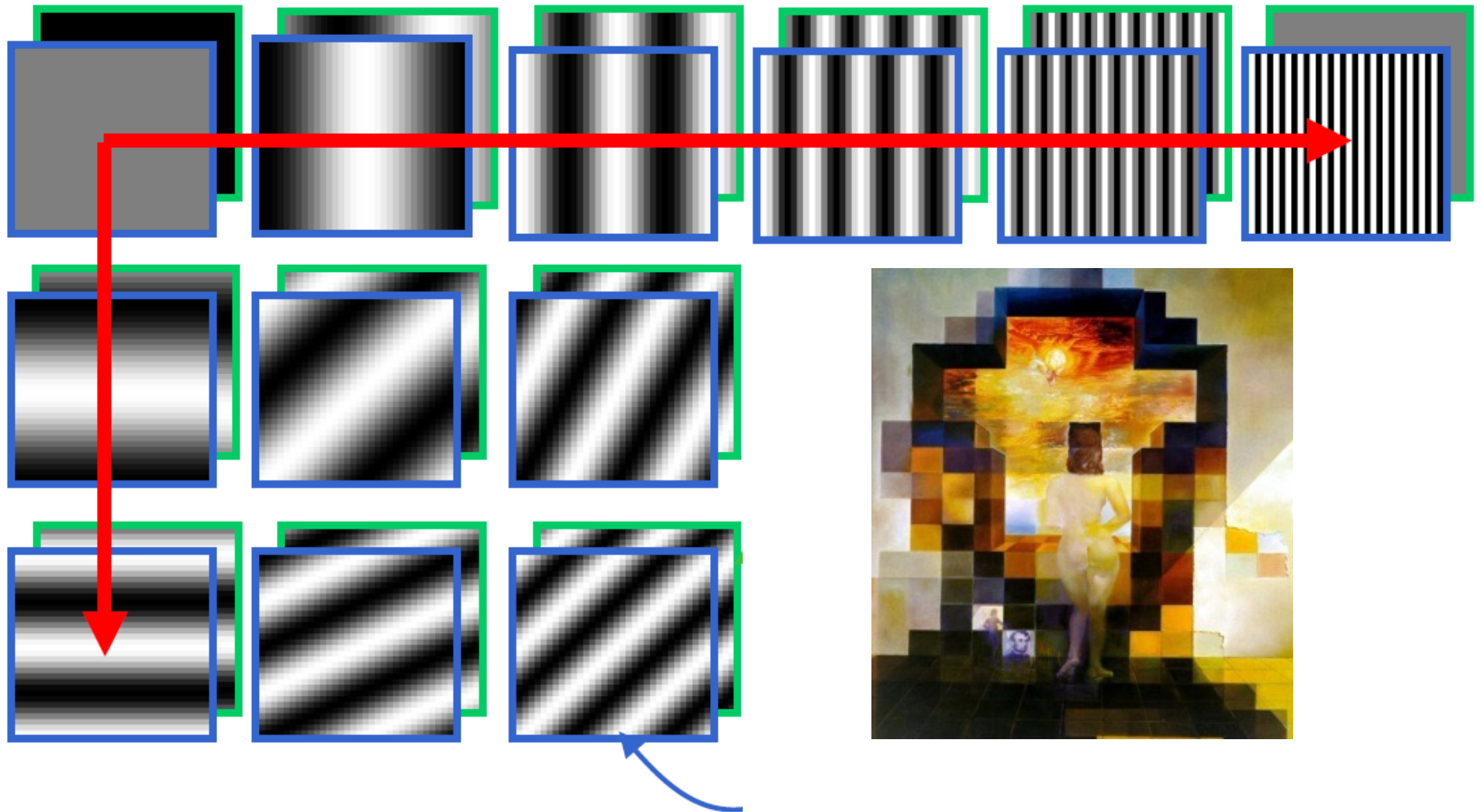
Spatial Frequencies and Perception



Campbell-Robson contrast sensitivity curve

A Nice Basis for 2D Image Functions

Teases away fast vs. slow changes in the image.



This change of basis has a special name...

Jean Baptiste Joseph Fourier (1768-1830)

had crazy idea (1807)

Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies.

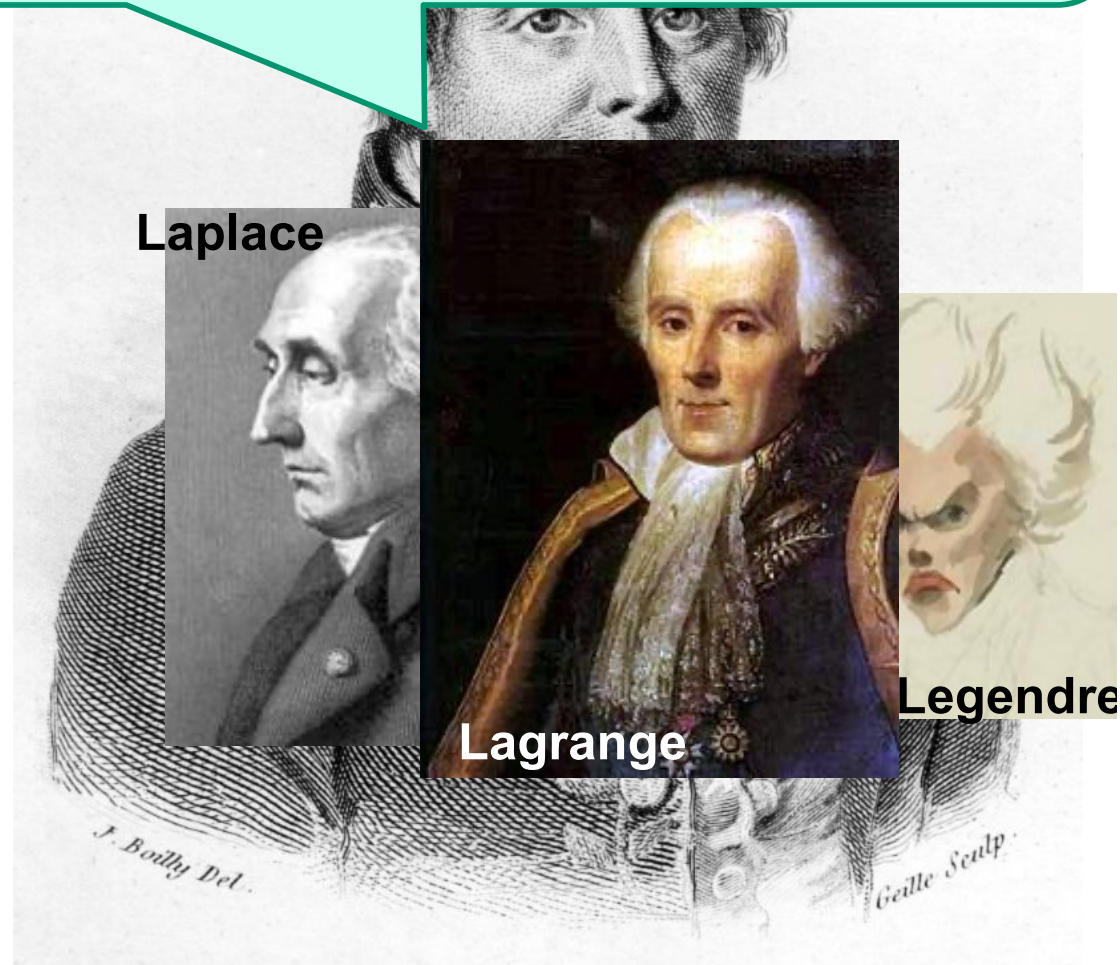
...the manner in which the author arrives at these equations is not exempt of difficulties and... his analysis to integrate them still leaves something to be desired on the score of generality and even rigour.

Don't believe it?

- Neither did Lagrange, Laplace, Poisson and other big wigs
- Not translated into English until 1878!

But it's (mostly) true!

- called Fourier Series



A sum of sines

Our building block:

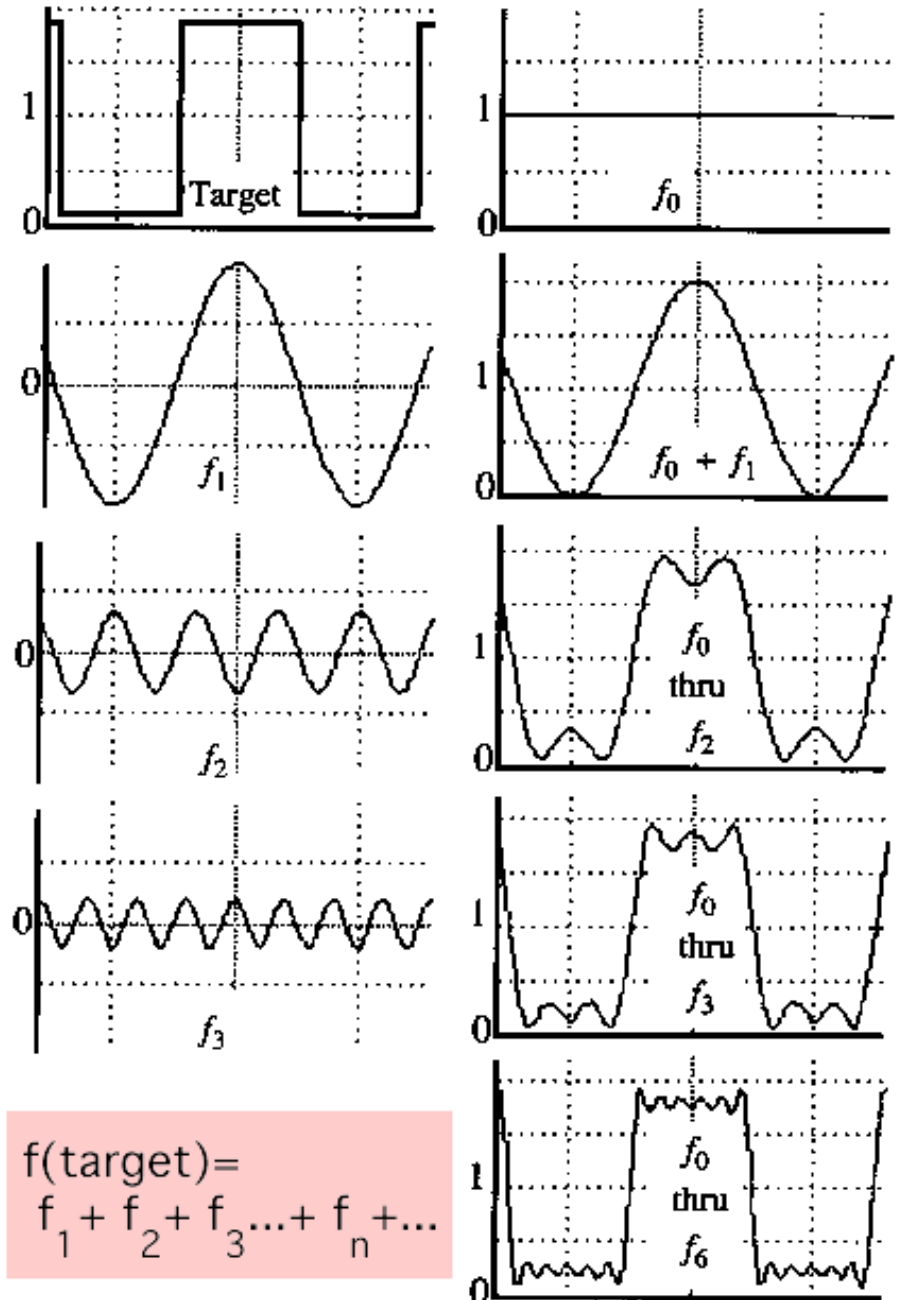
$$A \sin(\omega x + \phi)$$

Add enough of them to get any signal $f(x)$ you want!

How many degrees of freedom?

What does each control?

Which one encodes the coarse vs. fine structure of the signal?



Fourier Transform

We want to understand the frequency ω of our signal. So, let's reparametrize the signal by ω instead of x :



For every ω from 0 to ∞ , $F(\omega)$ holds the amplitude A and phase ϕ of the corresponding sine $A \sin(\omega x + \phi)$

- How does F hold both?

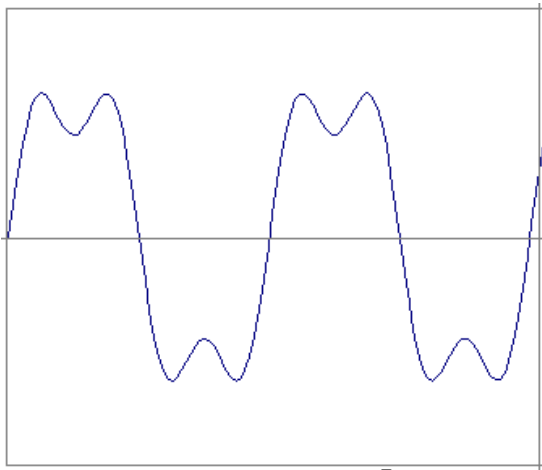
$$F(\omega) = R(\omega) + iI(\omega)$$
$$A = \pm \sqrt{R(\omega)^2 + I(\omega)^2} \qquad \phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$

We can always go back:



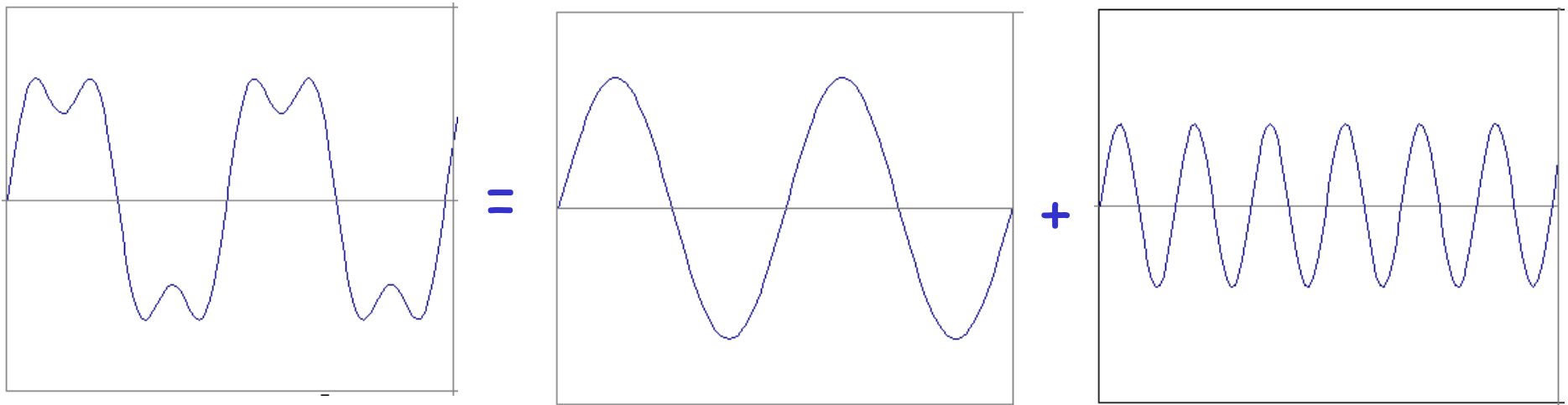
Time and Frequency

example : $g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi (3f) t)$



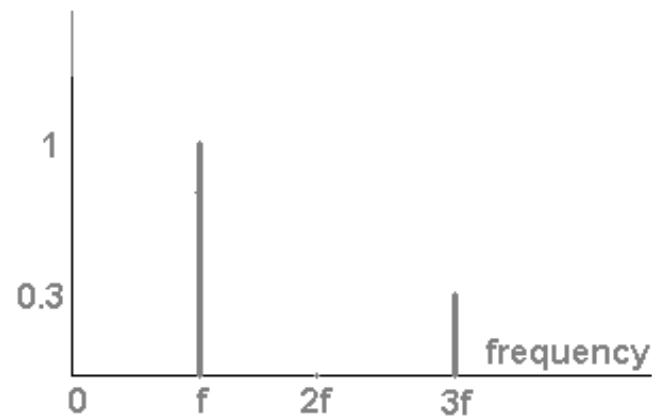
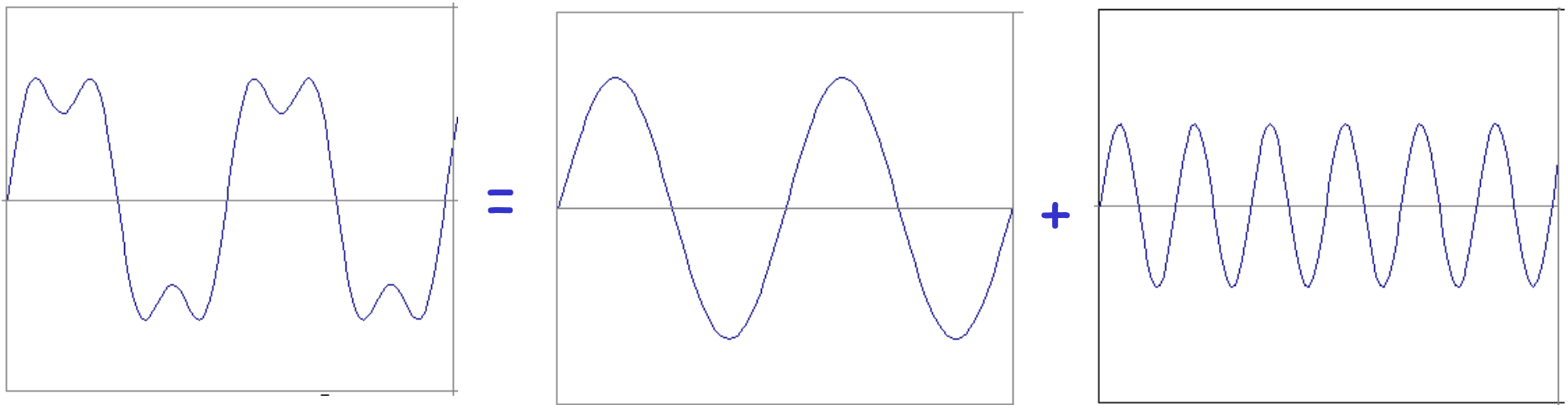
Time and Frequency

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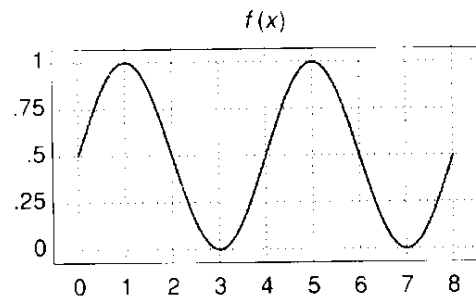


Frequency Spectrum

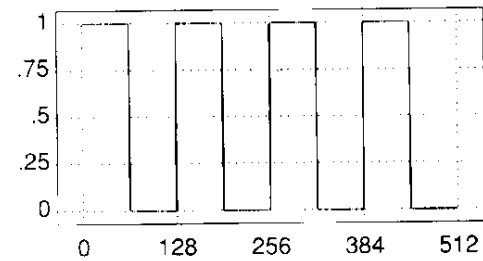
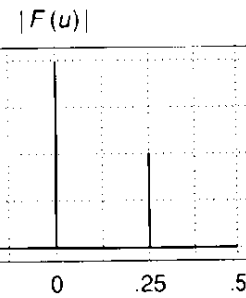
example : $g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi (3f) t)$



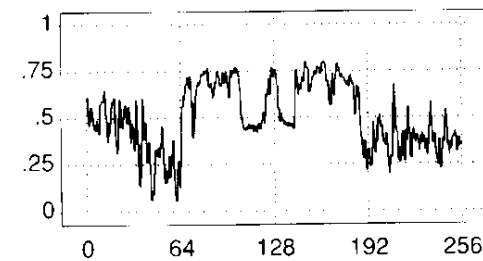
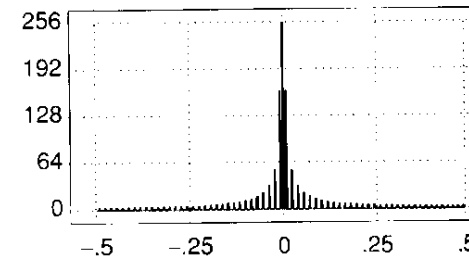
Various Frequency Spectra



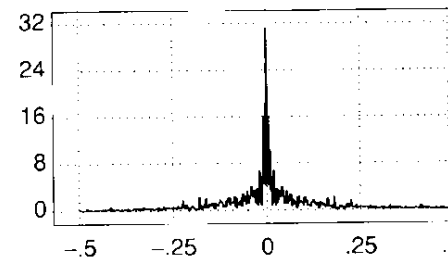
(a)



(b)

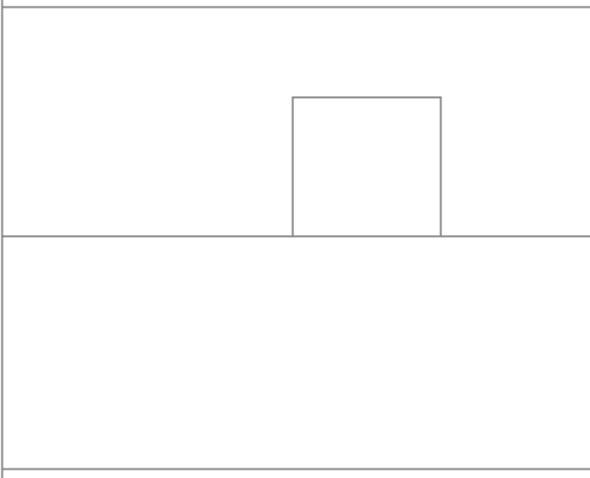


(c)

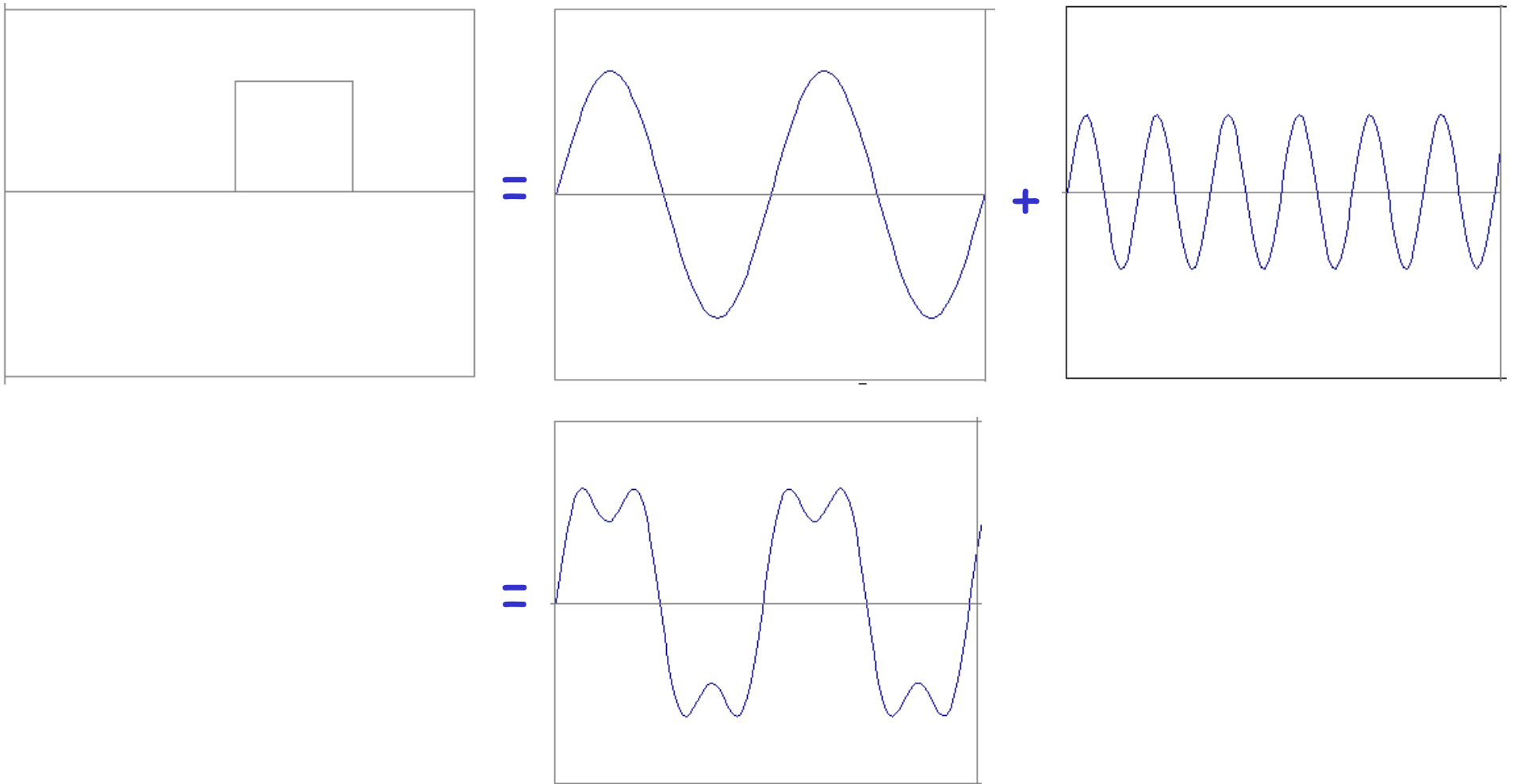


Frequency Spectra

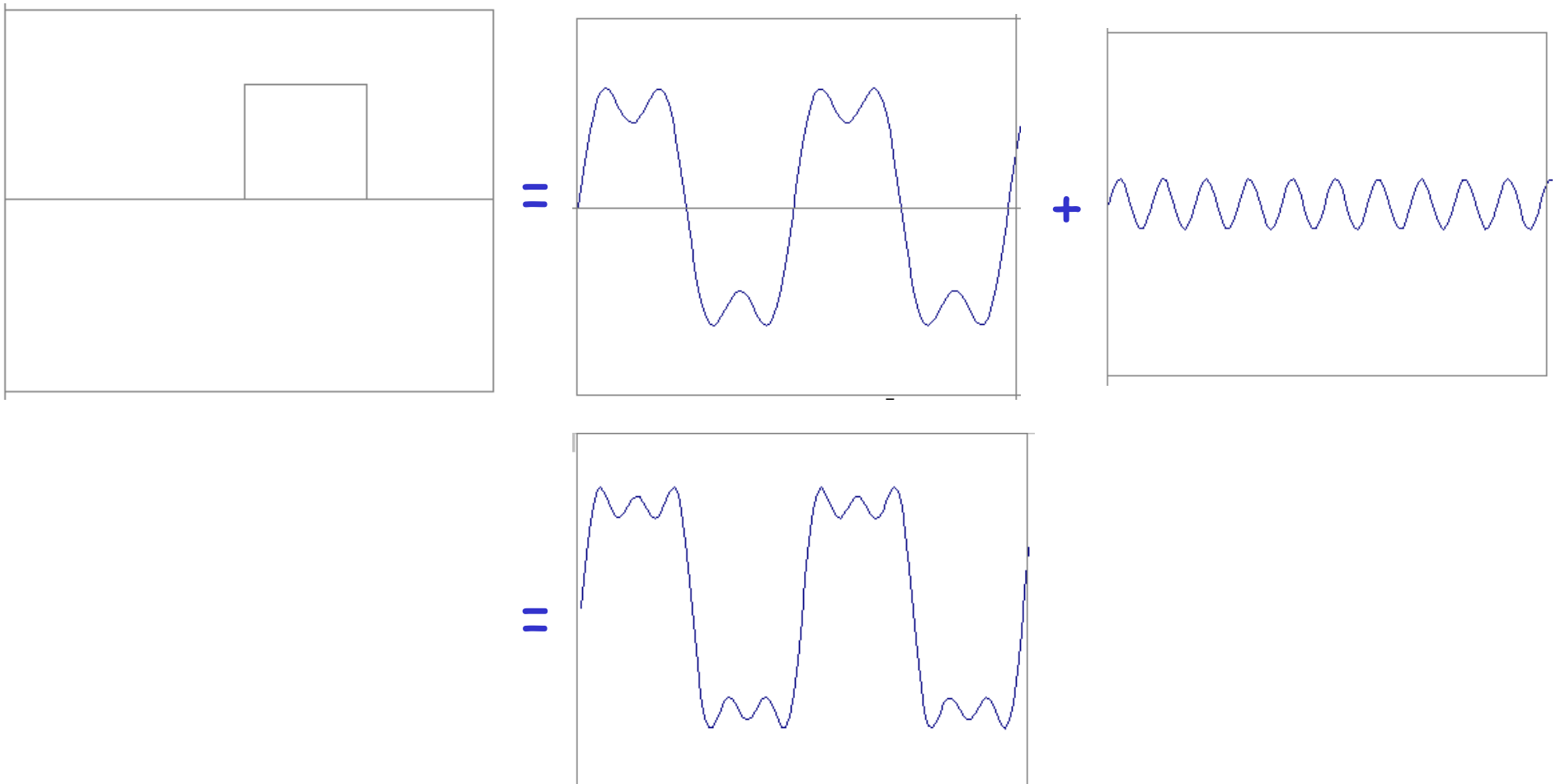
Usually, frequency is more interesting than the phase



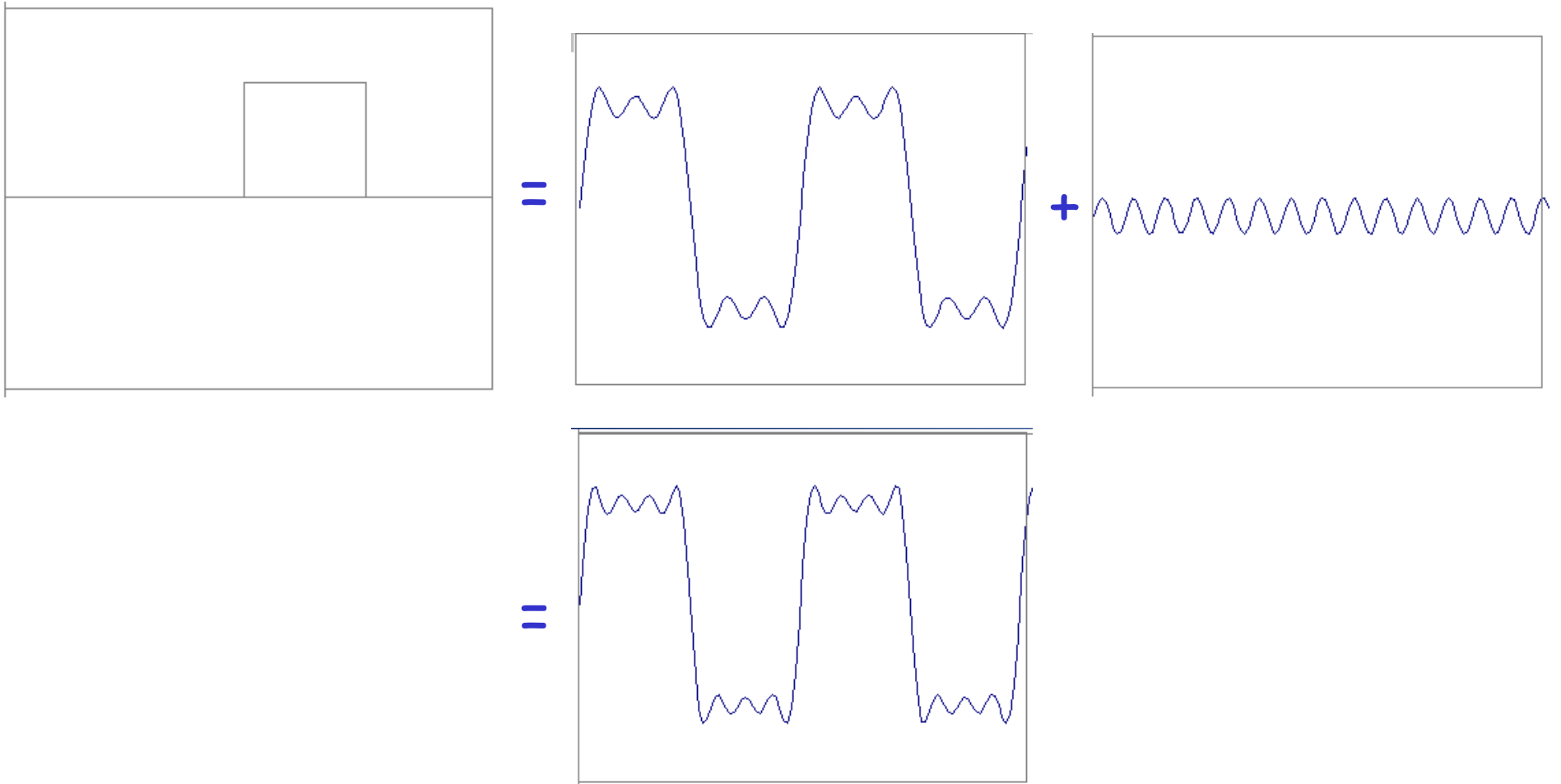
Frequency Spectra



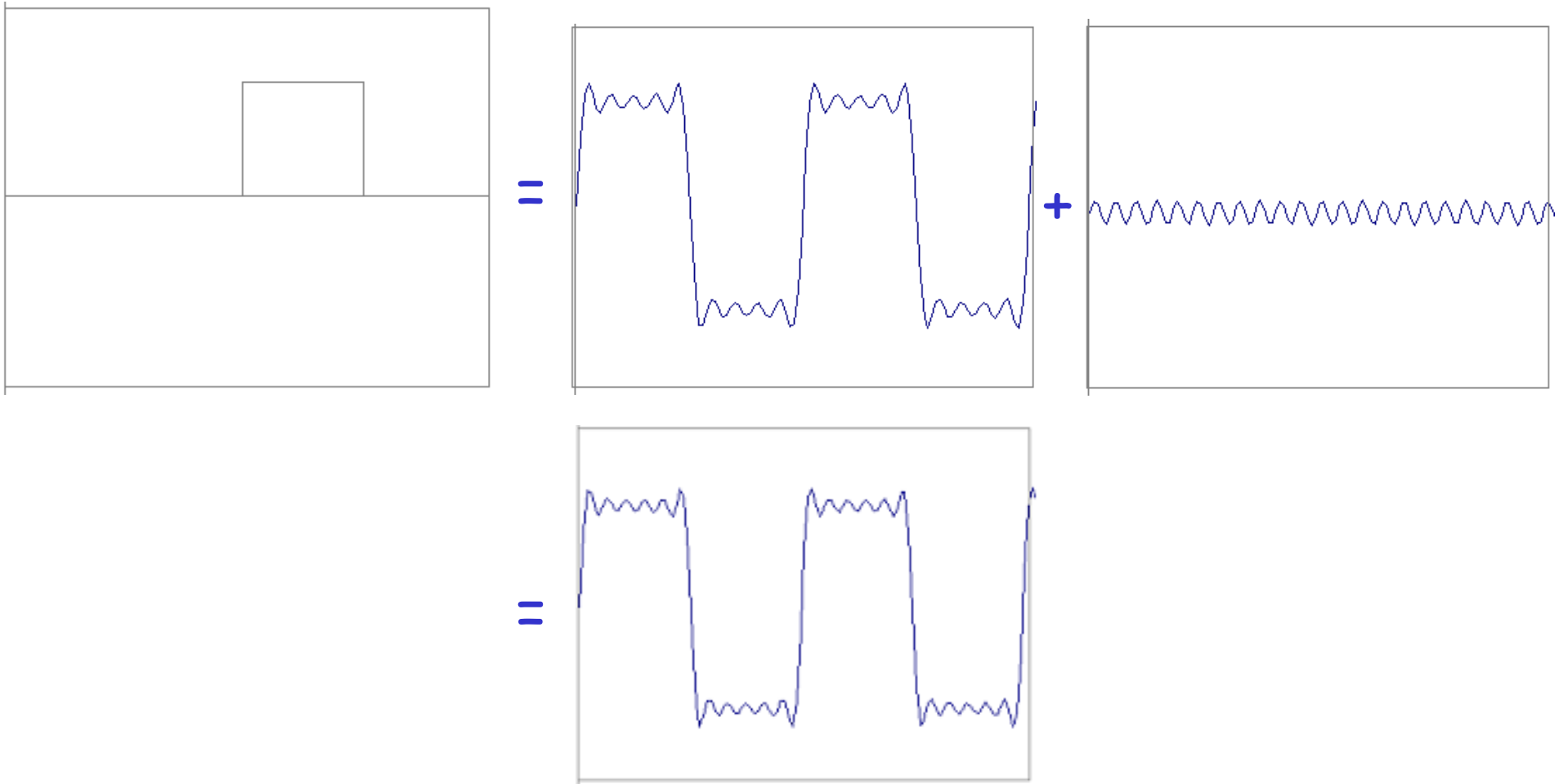
Frequency Spectra



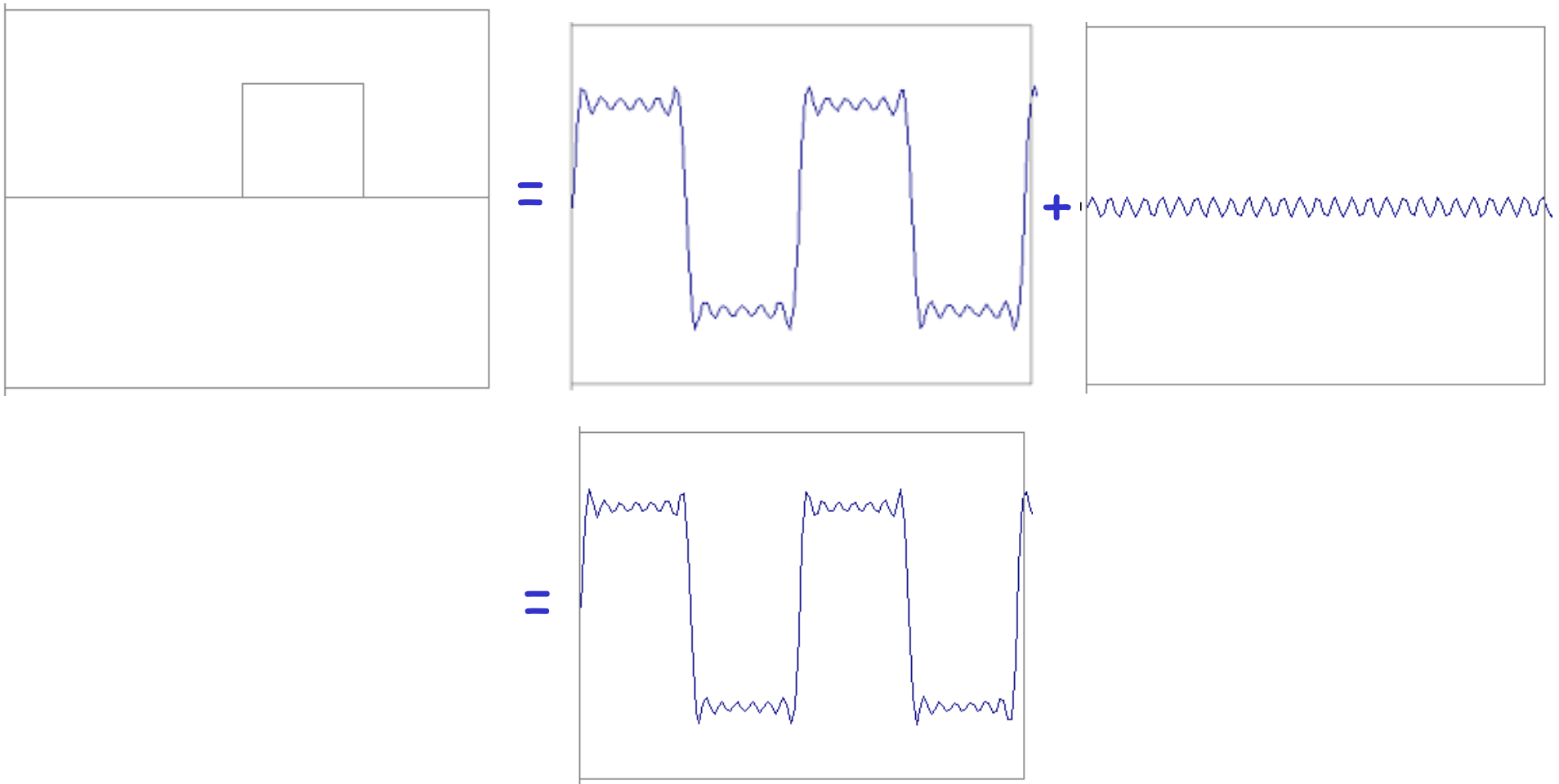
Frequency Spectra



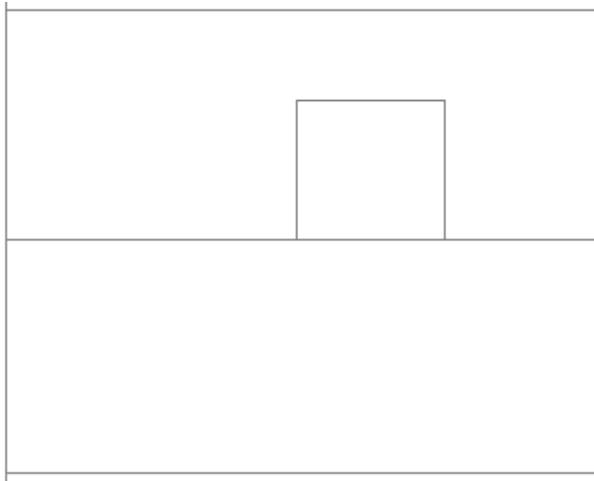
Frequency Spectra



Frequency Spectra

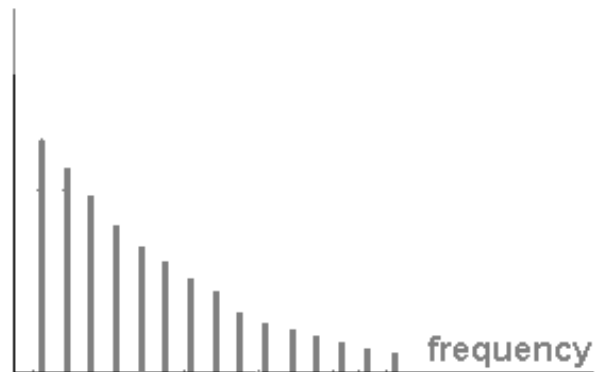


Frequency Spectra

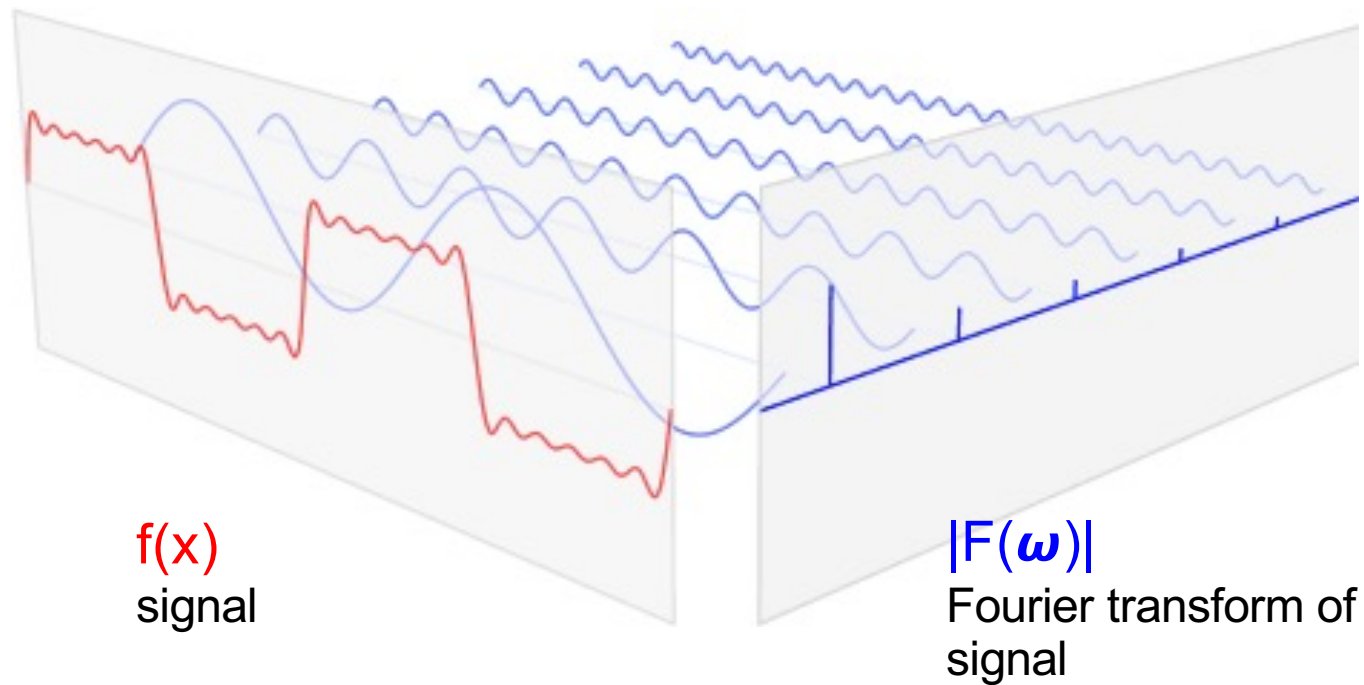


=

$$A \sum_{k=1}^{\infty} \frac{1}{k} \sin(2\pi kt)$$

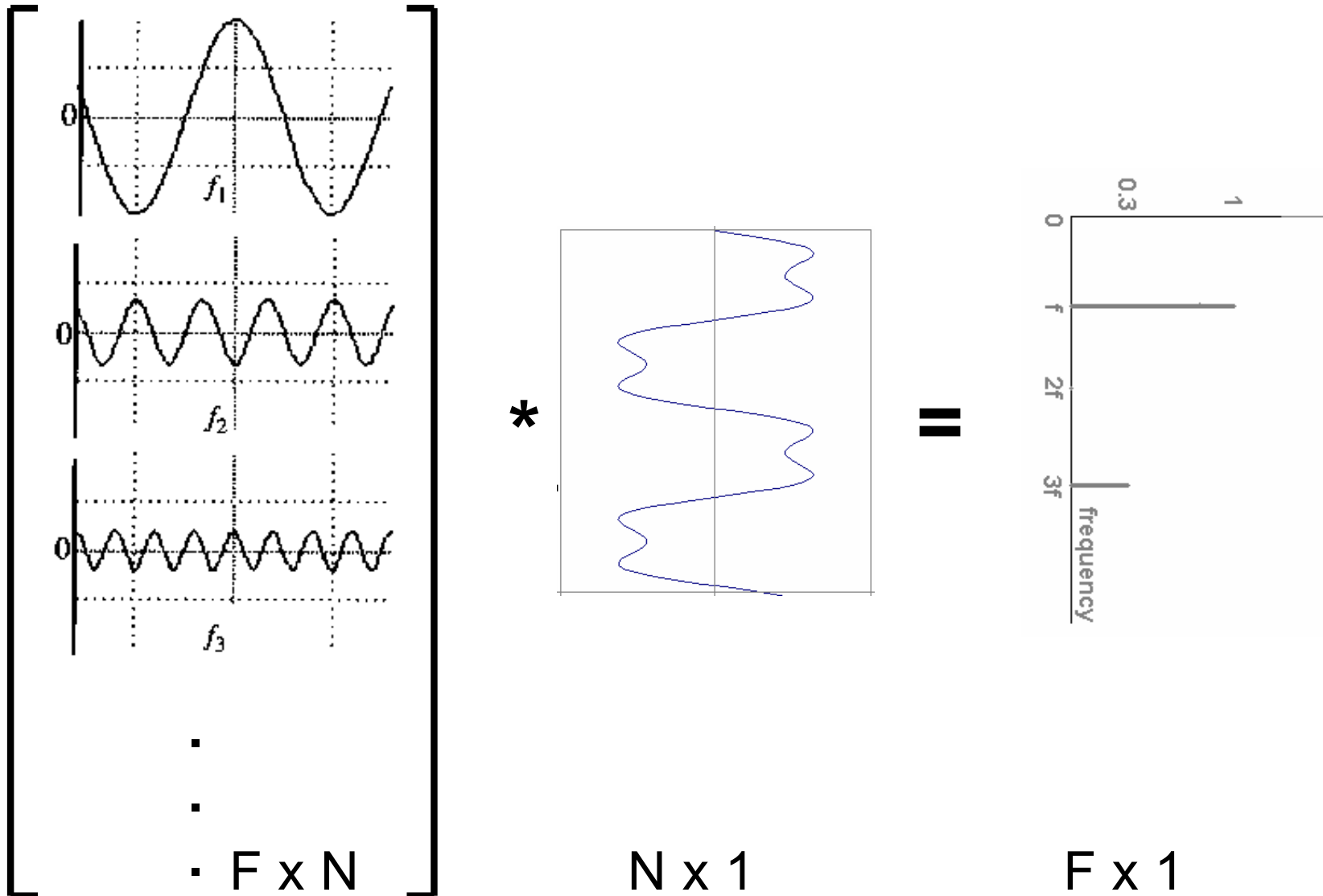


Signal and its Furrier Transform



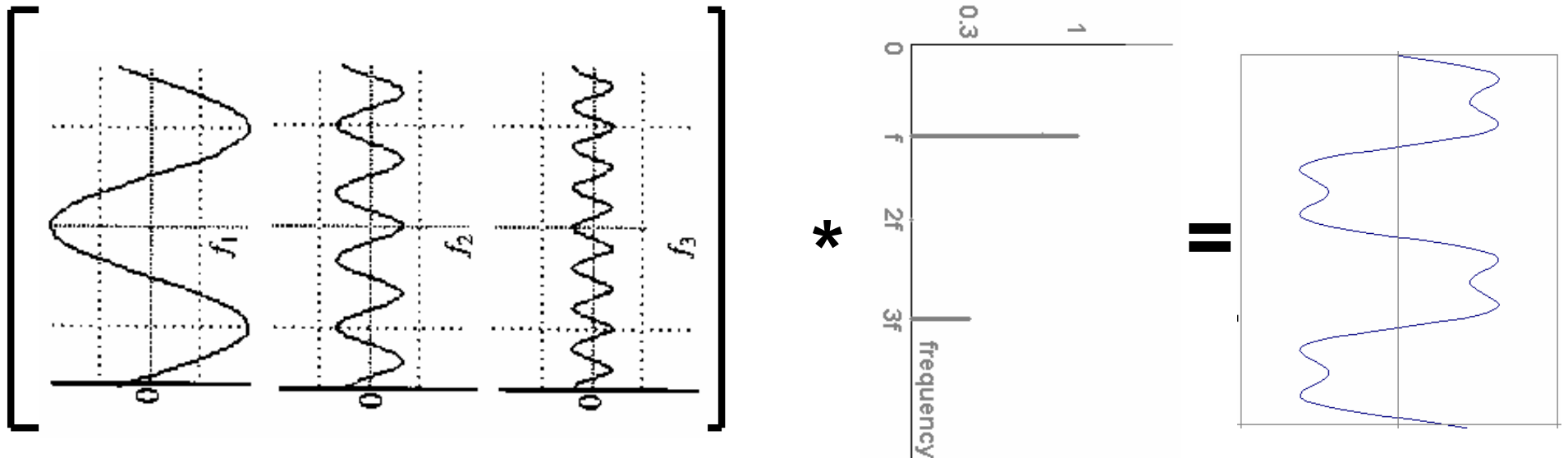
FT: Just a change of basis

$$M * f(x) = F(\omega)$$



IFT: Just a change of basis

$$M^{-1} * F(\omega) = f(x)$$



•

•

• $N \times F$

$F \times 1$

$N \times 1$

Finally: Scary Math

$$\text{Fourier Transform : } F(\omega) = \int_{-\infty}^{+\infty} f(x)e^{-i\omega x} dx$$

$$\text{Inverse Fourier Transform : } f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega)e^{i\omega x} d\omega$$

Finally: Scary Math

$$\text{Fourier Transform : } F(\omega) = \int_{-\infty}^{+\infty} f(x)e^{-i\omega x} dx$$

$$\text{Inverse Fourier Transform : } f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega)e^{i\omega x} d\omega$$

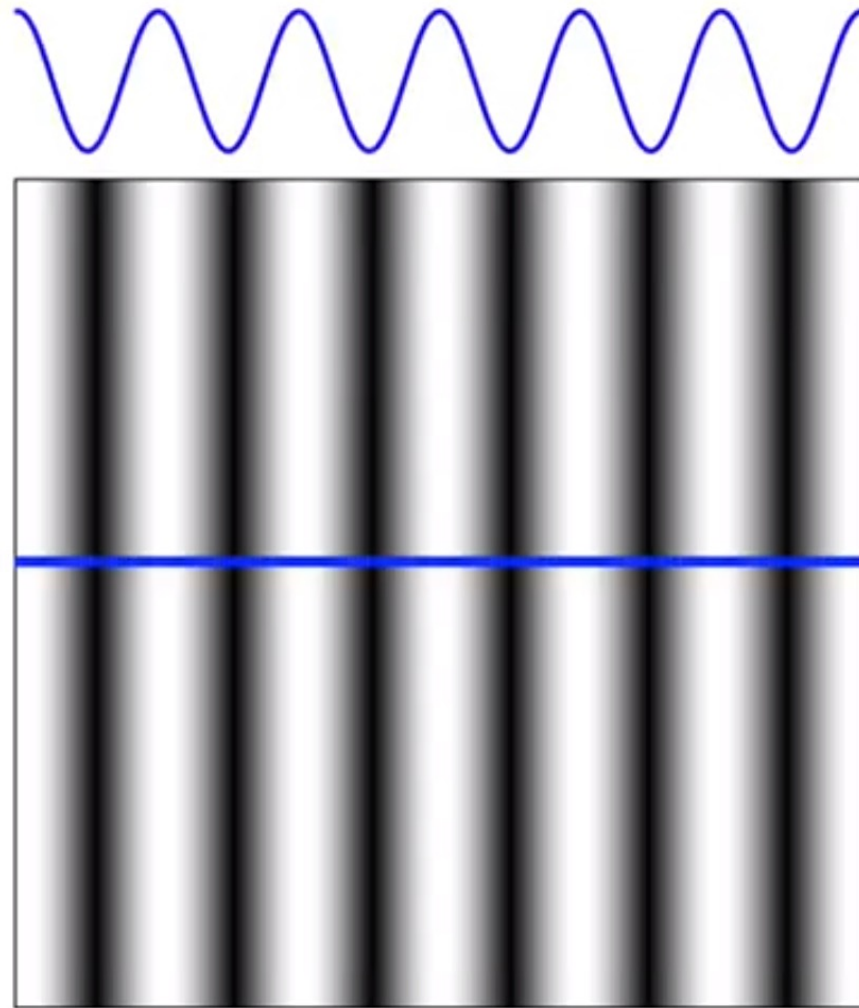
...not really scary: $e^{i\omega x} = \cos(\omega x) + i \sin(\omega x)$

is hiding our old friend: $\sin(\omega x + \phi)$

$$\begin{array}{l} \text{phase can be encoded} \\ \text{by sin/cos pair} \end{array} \rightarrow \begin{array}{l} P \cos(x) + Q \sin(x) = A \sin(x + \phi) \\ A = \pm \sqrt{P^2 + Q^2} \quad \phi = \tan^{-1}\left(\frac{P}{Q}\right) \end{array}$$

So it's just our signal $f(x)$ times sine at frequency ω

Extension to 2D



Extension to 2D

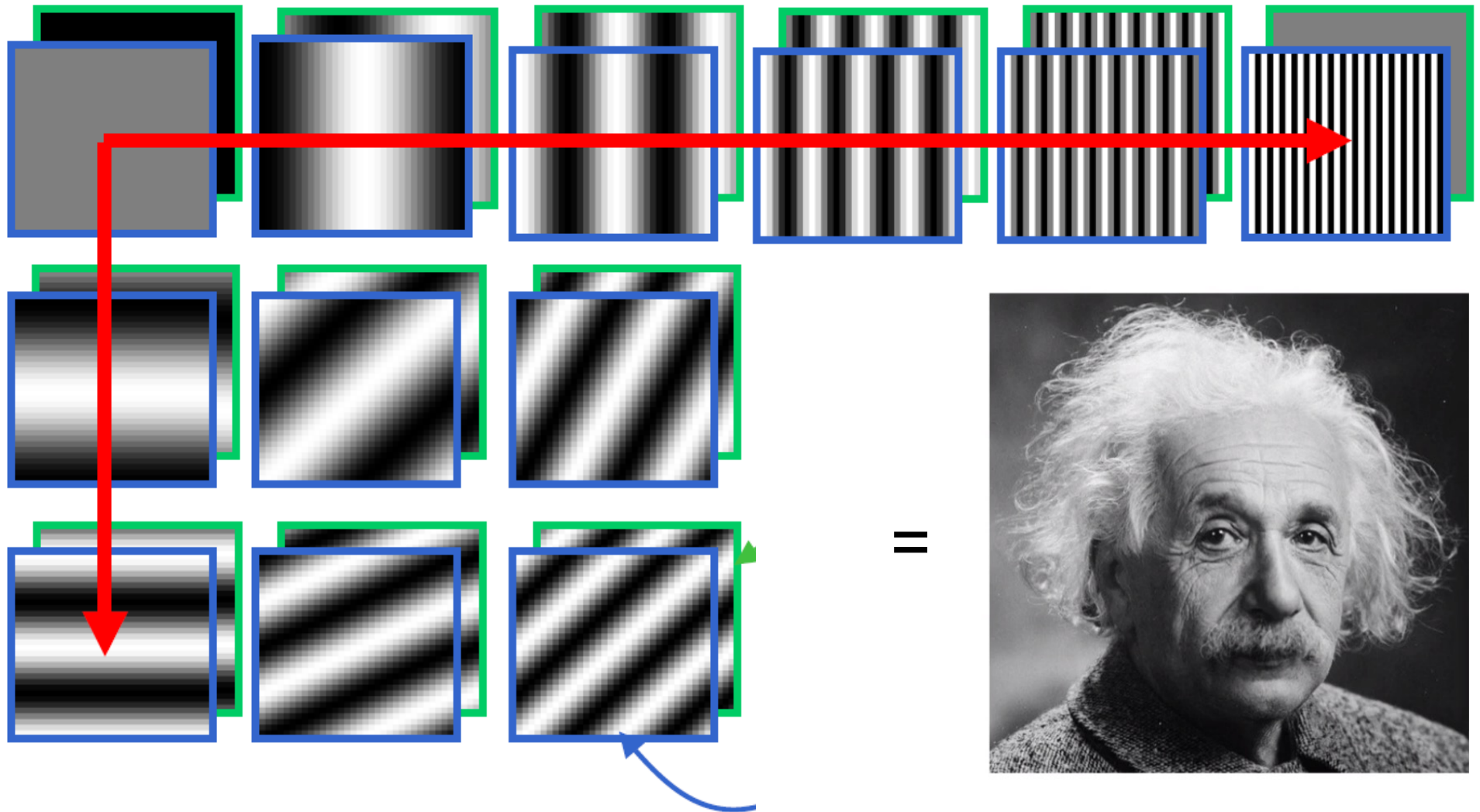
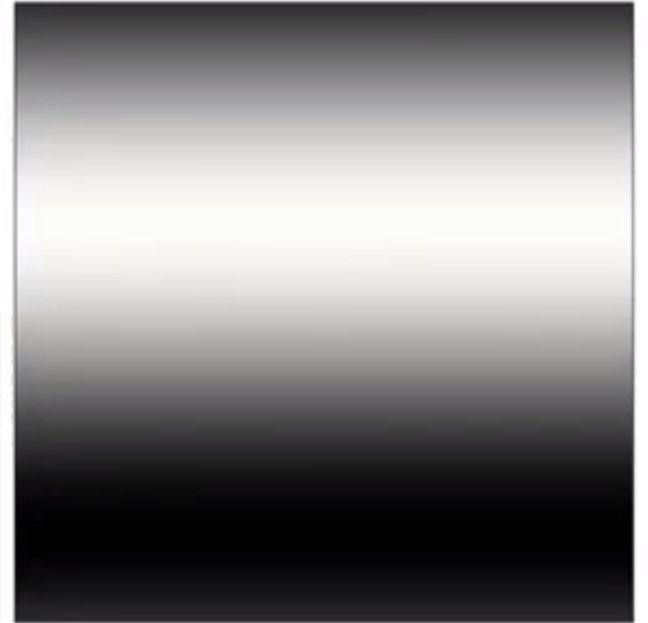
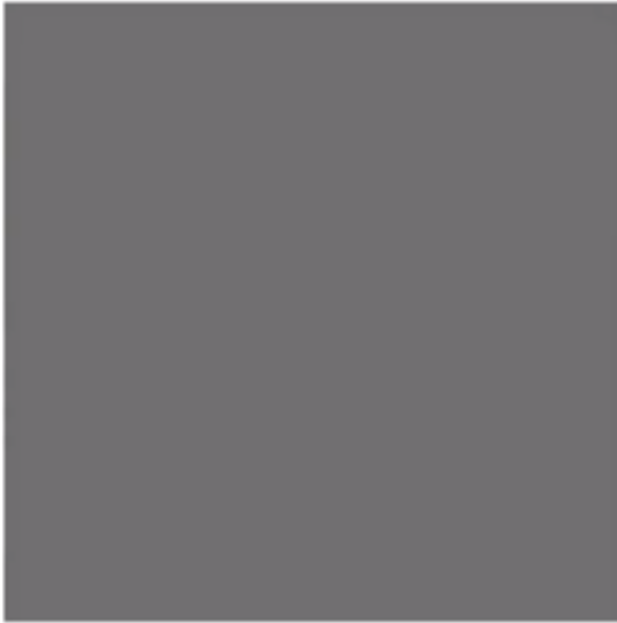
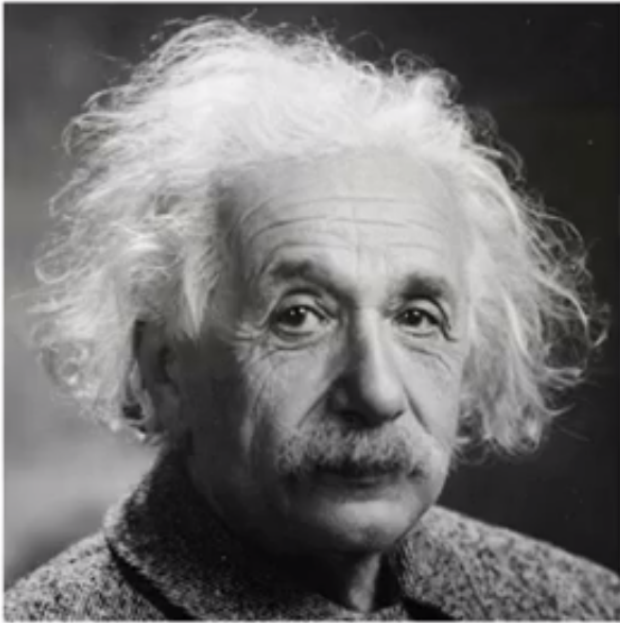
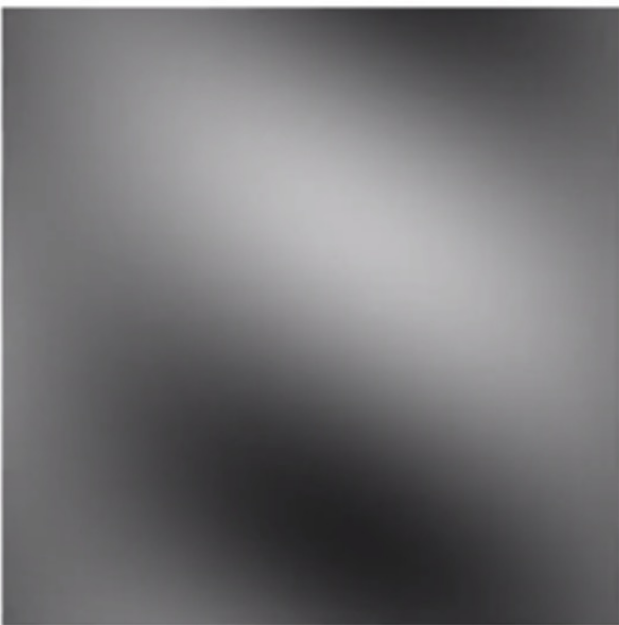
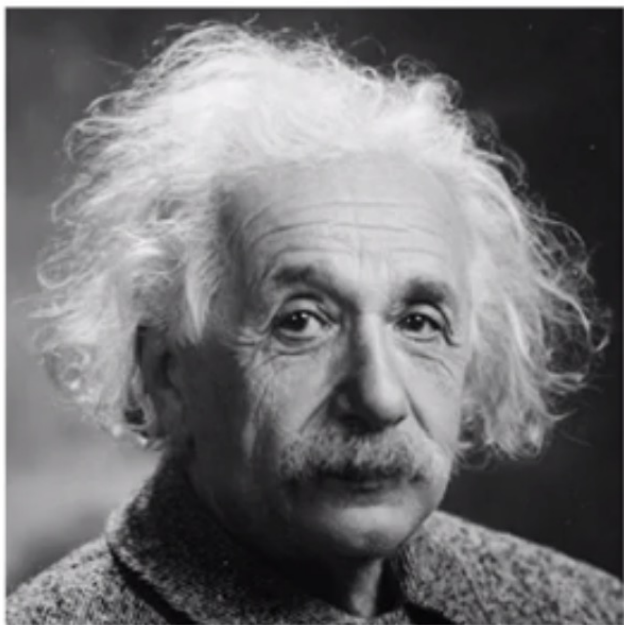
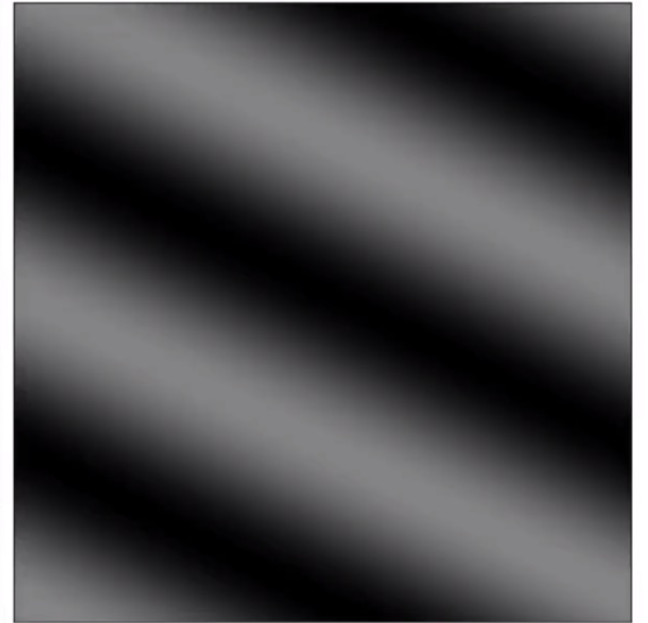
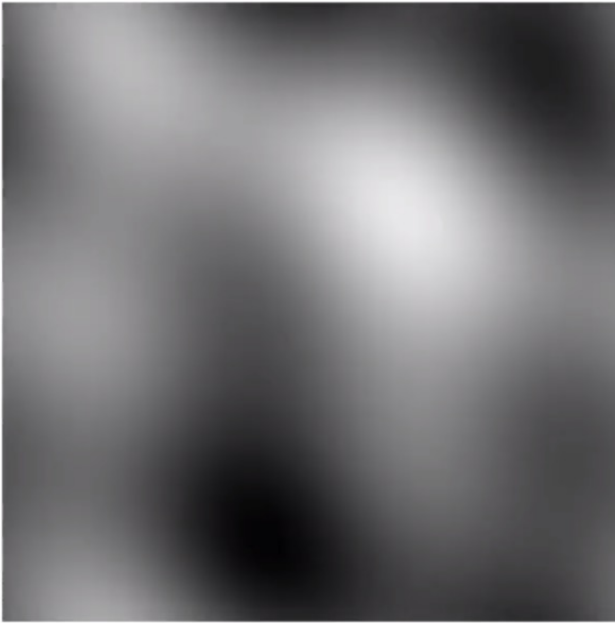
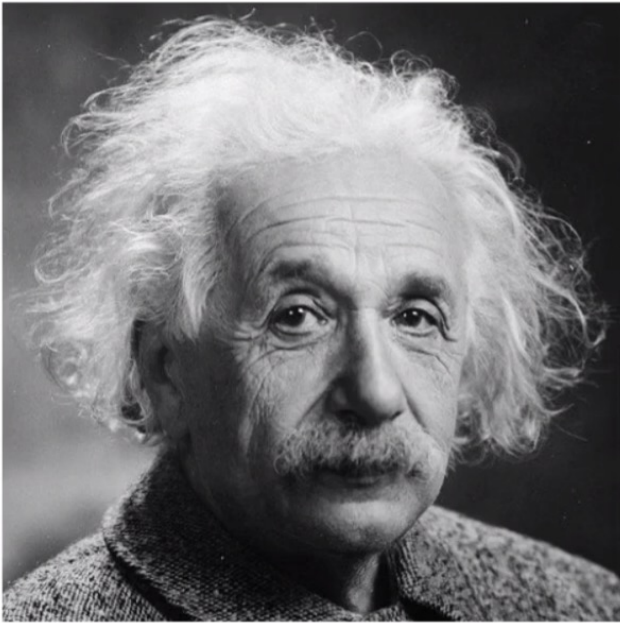


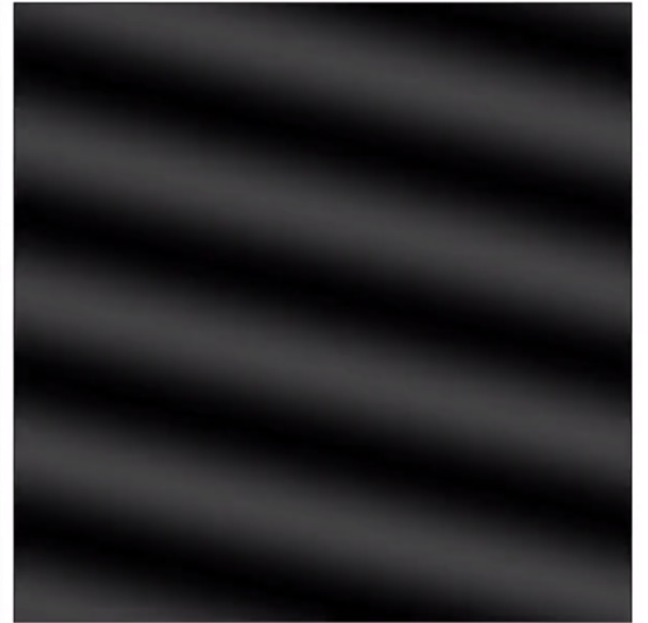
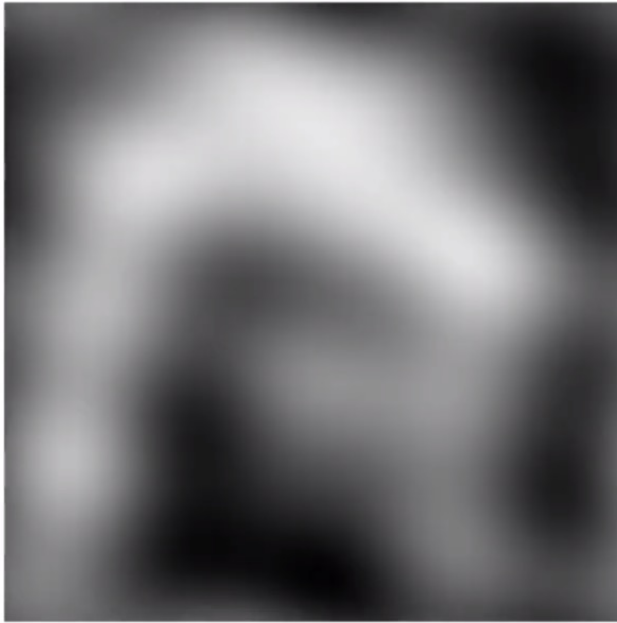
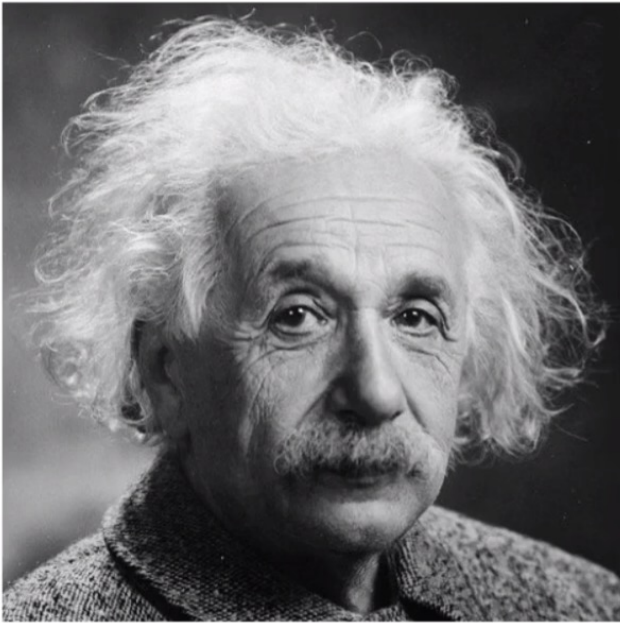
Image as a sum of basis images

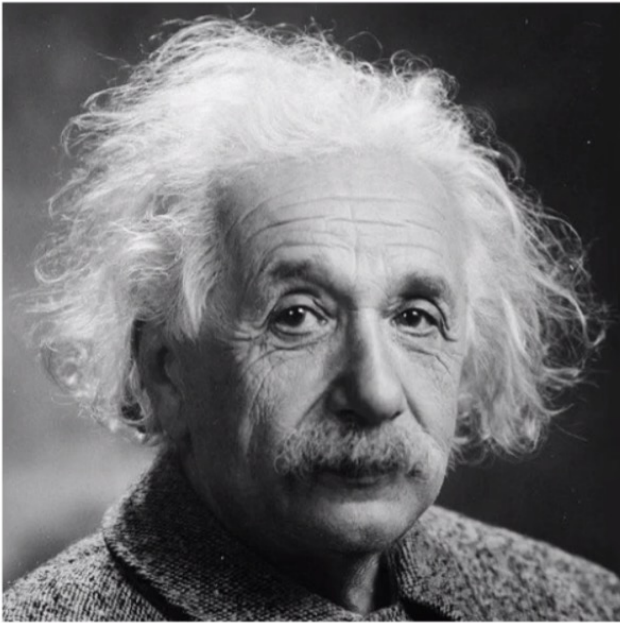


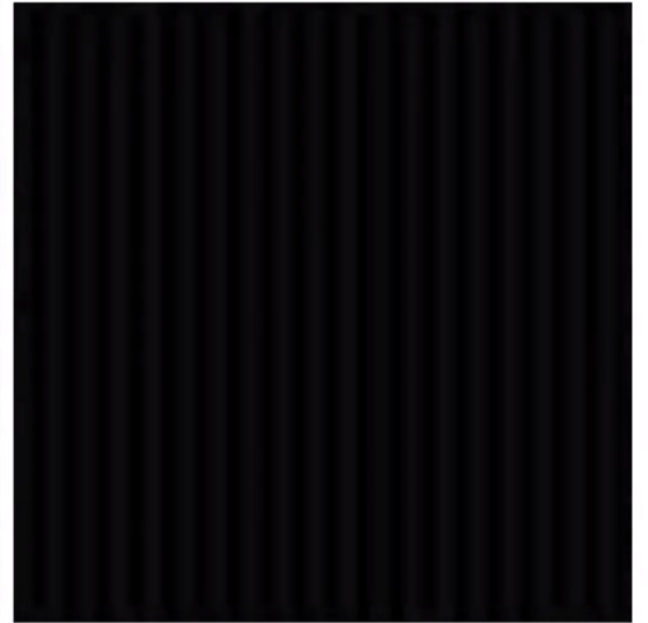
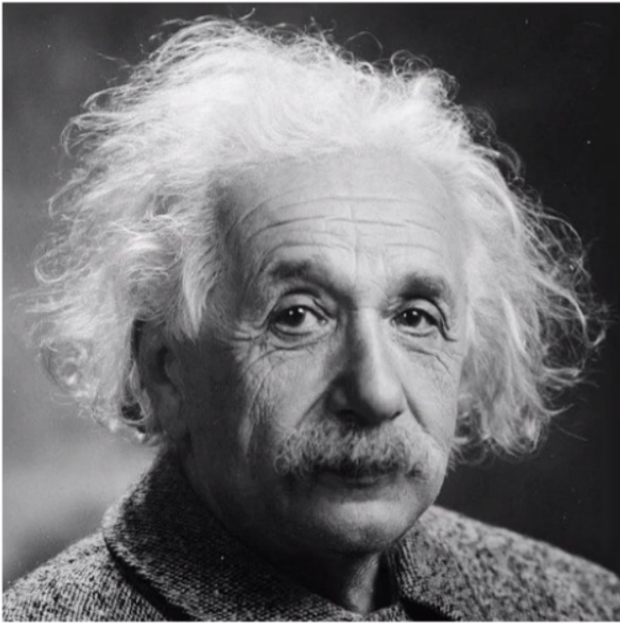
Contrast x3









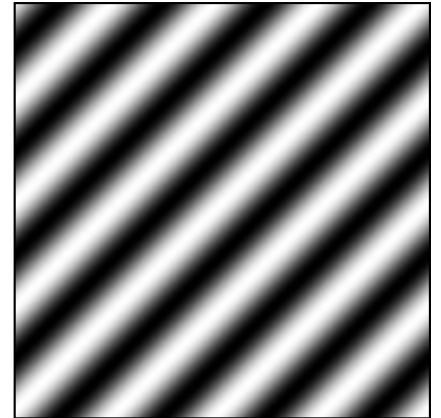
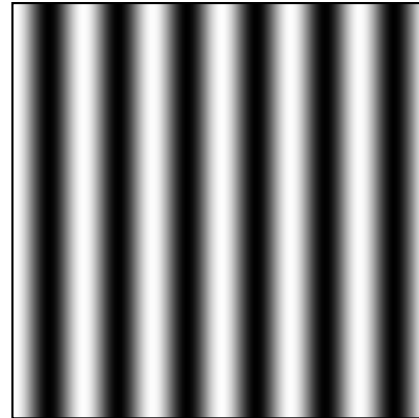
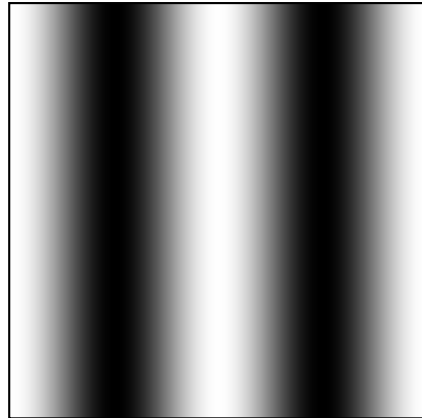


100

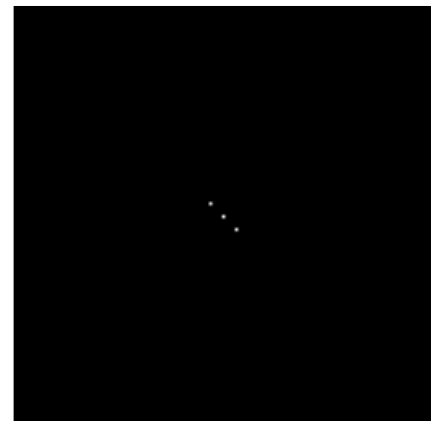
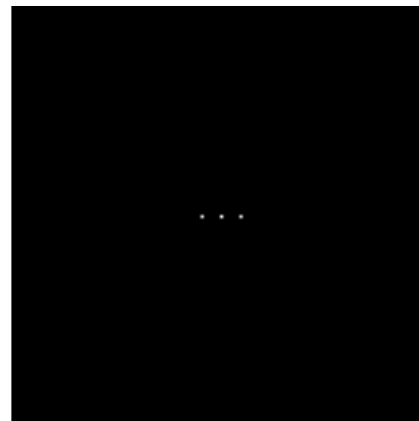
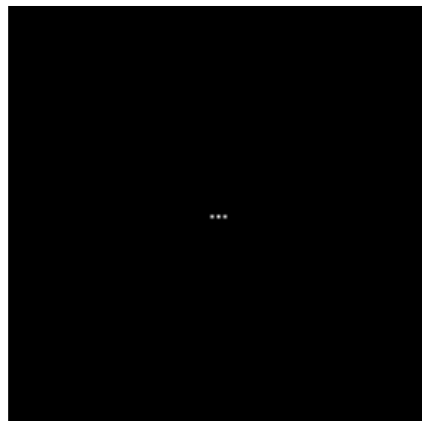


Fourier analysis in images

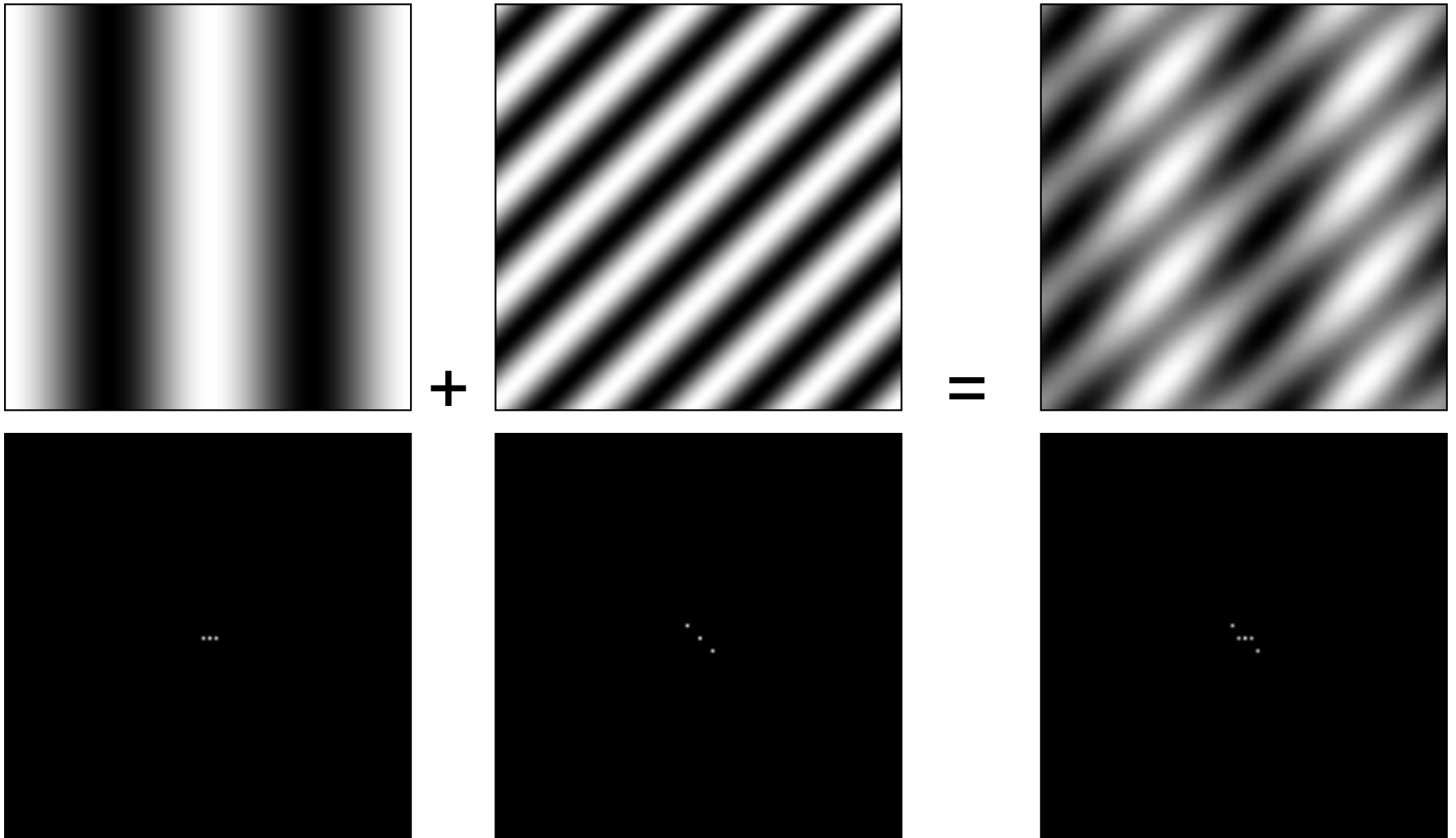
Intensity Image



Fourier Image



Signals can be composed

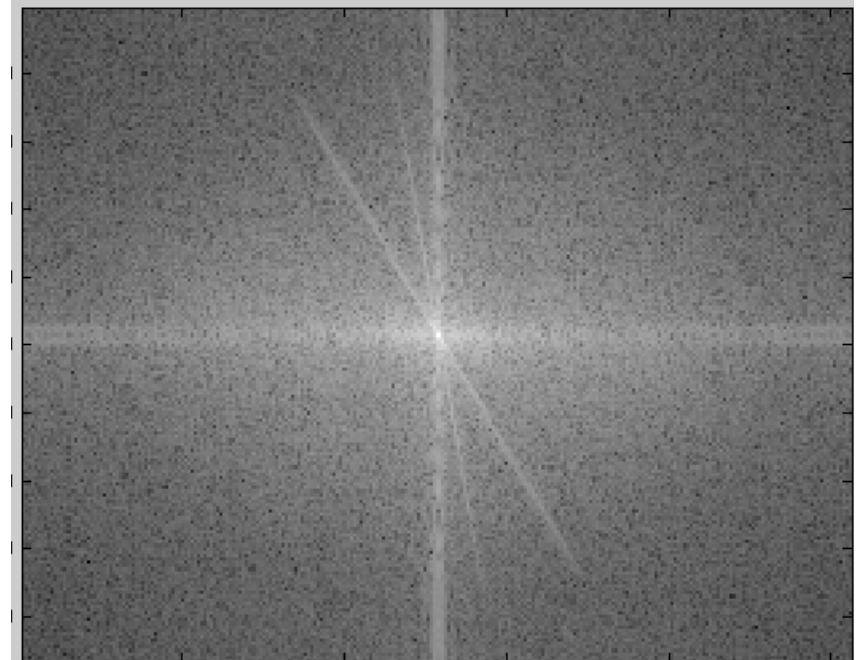


<http://sharp.bu.edu/~slehar/fourier/fourier.html#filtering>
More: <http://www.cs.unm.edu/~brayer/vision/fourier.html>

Architectural Scene



Amplitude Spectrum

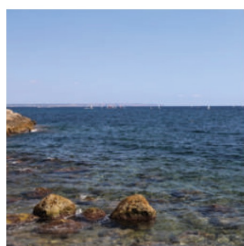


Can change spectrum, then reconstruct



Local change in one domain, causes global change in the other

The Furrier Game: find the right pairs



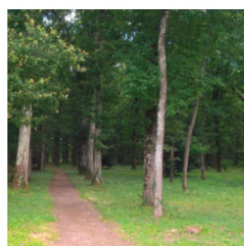
a)



b)



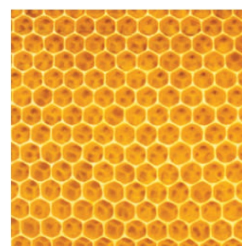
c)



d)



e)



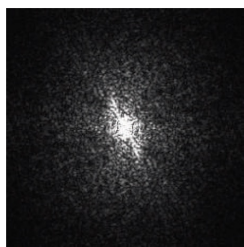
f)



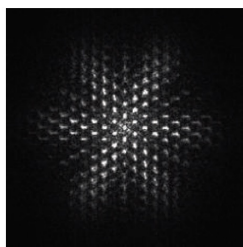
g)



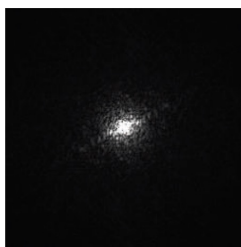
h)



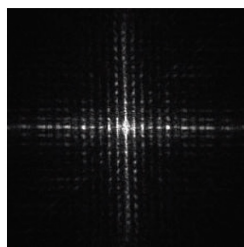
1)



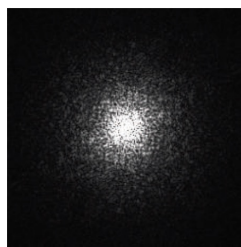
2)



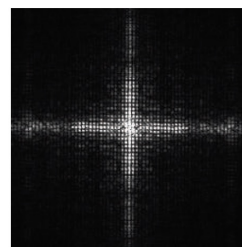
3)



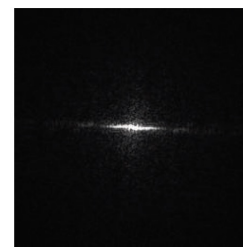
4)



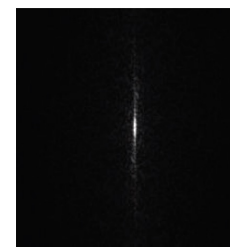
5)



6)



7)

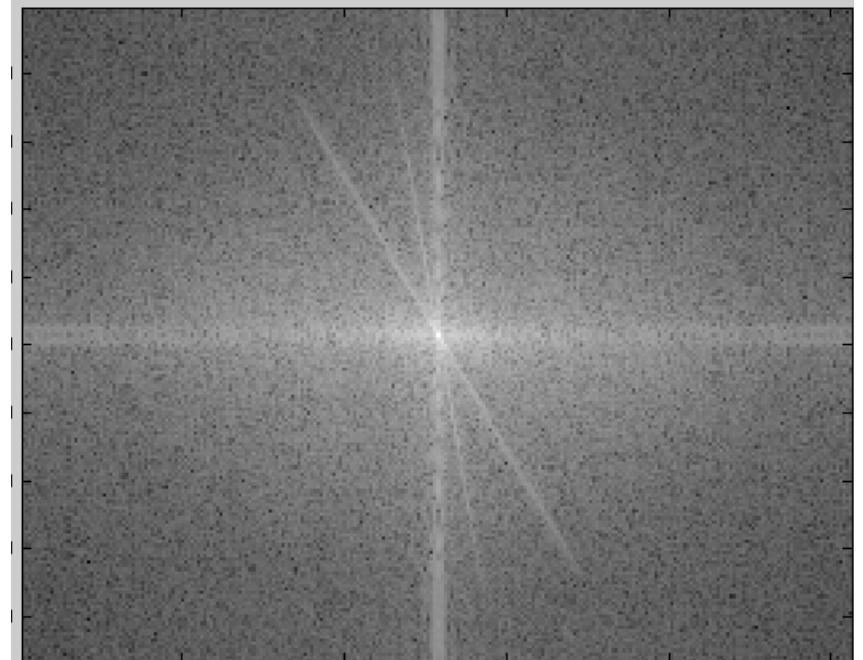


8)

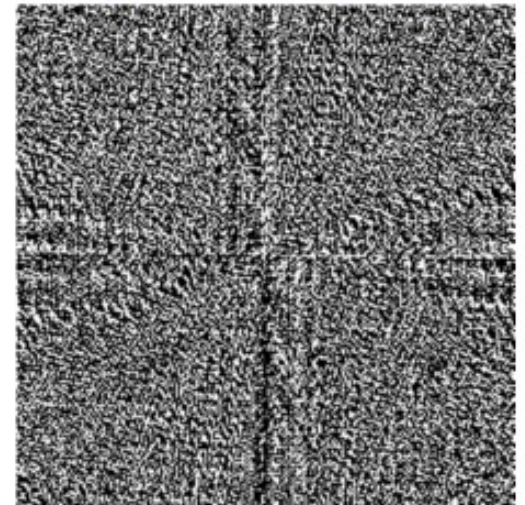
What about phase?



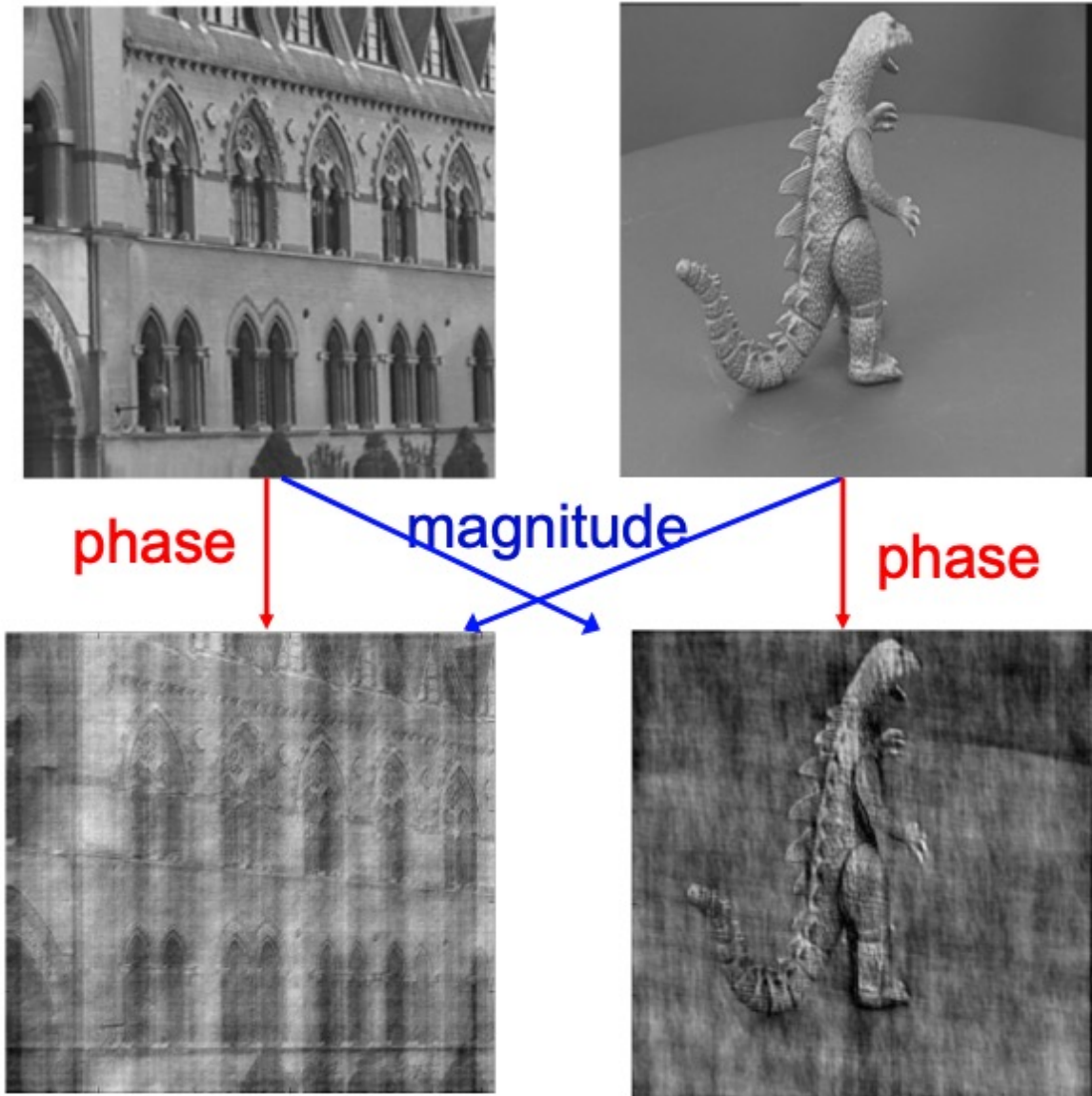
Amplitude Spectrum



what does phase look like, you ask?
(less visually informative)

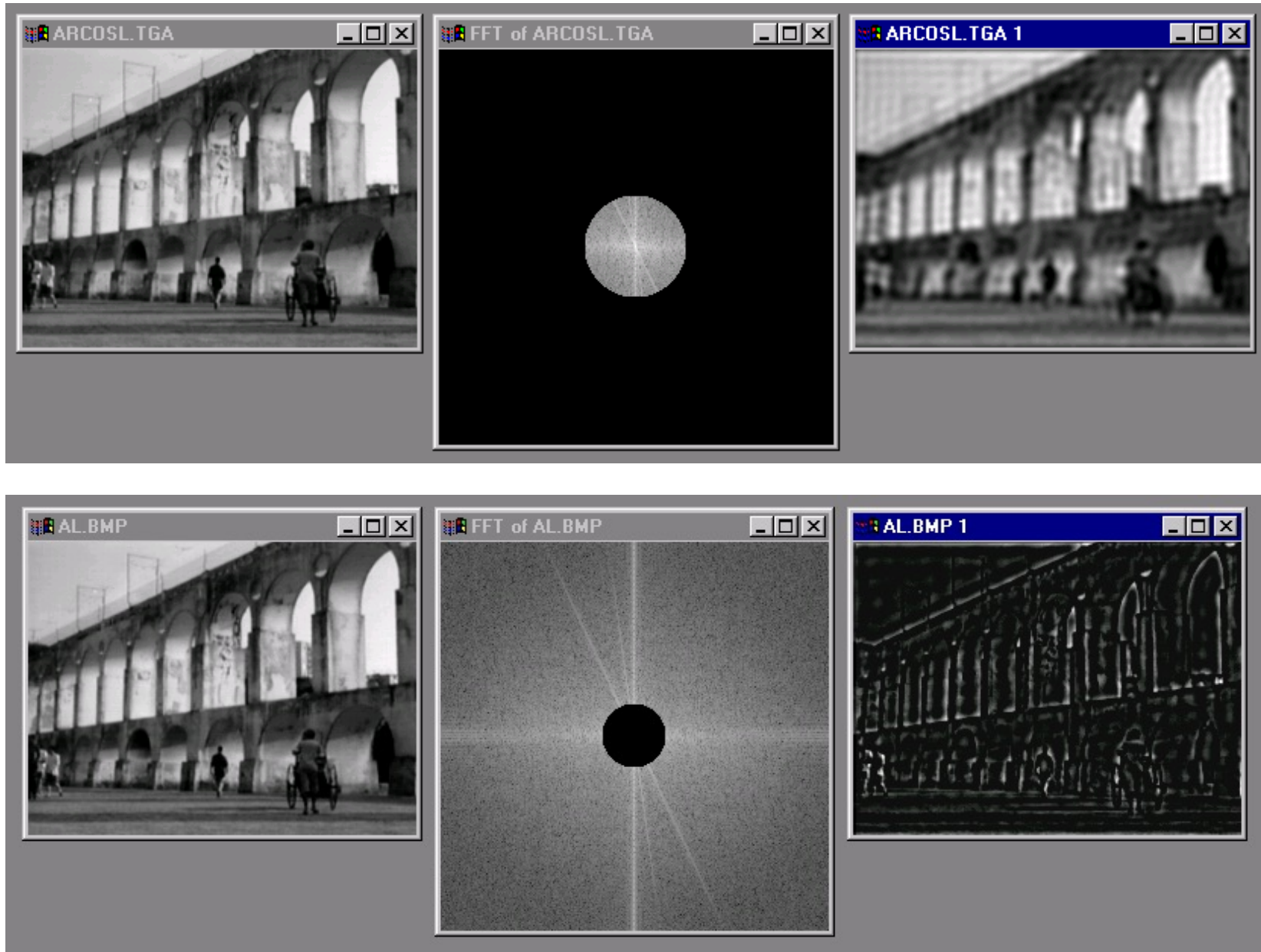


The importance of Phase



Slide: Andrew Zisserman

Low and High Pass filtering



The Convolution Theorem

The greatest thing since sliced (banana) bread!

- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g]F[h]$$

- The inverse Fourier transform of the product of two Fourier transforms is the convolution of the two inverse Fourier transforms

$$F^{-1}[gh] = F^{-1}[g] * F^{-1}[h]$$

- **Convolution** in spatial domain is equivalent to **multiplication** in frequency domain!

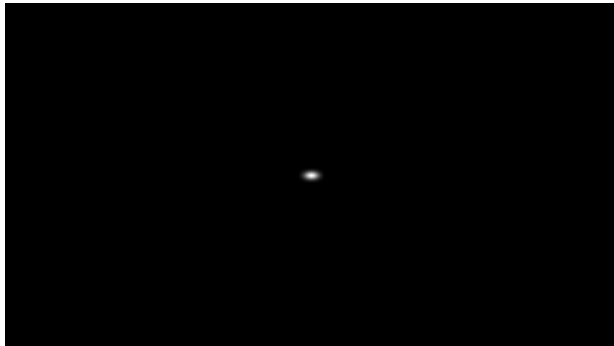
2D convolution theorem example

$f(x,y)$



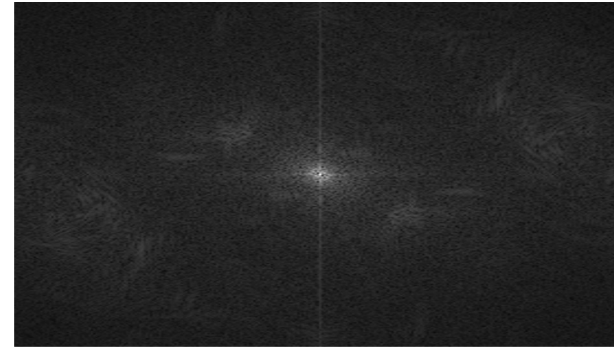
*

$h(x,y)$



$\Downarrow$

$g(x,y)$



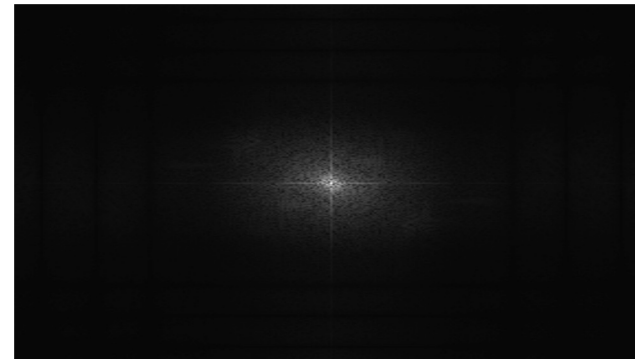
$\times$

$|F(s_x, s_y)|$



$\Downarrow$

$|H(s_x, s_y)|$

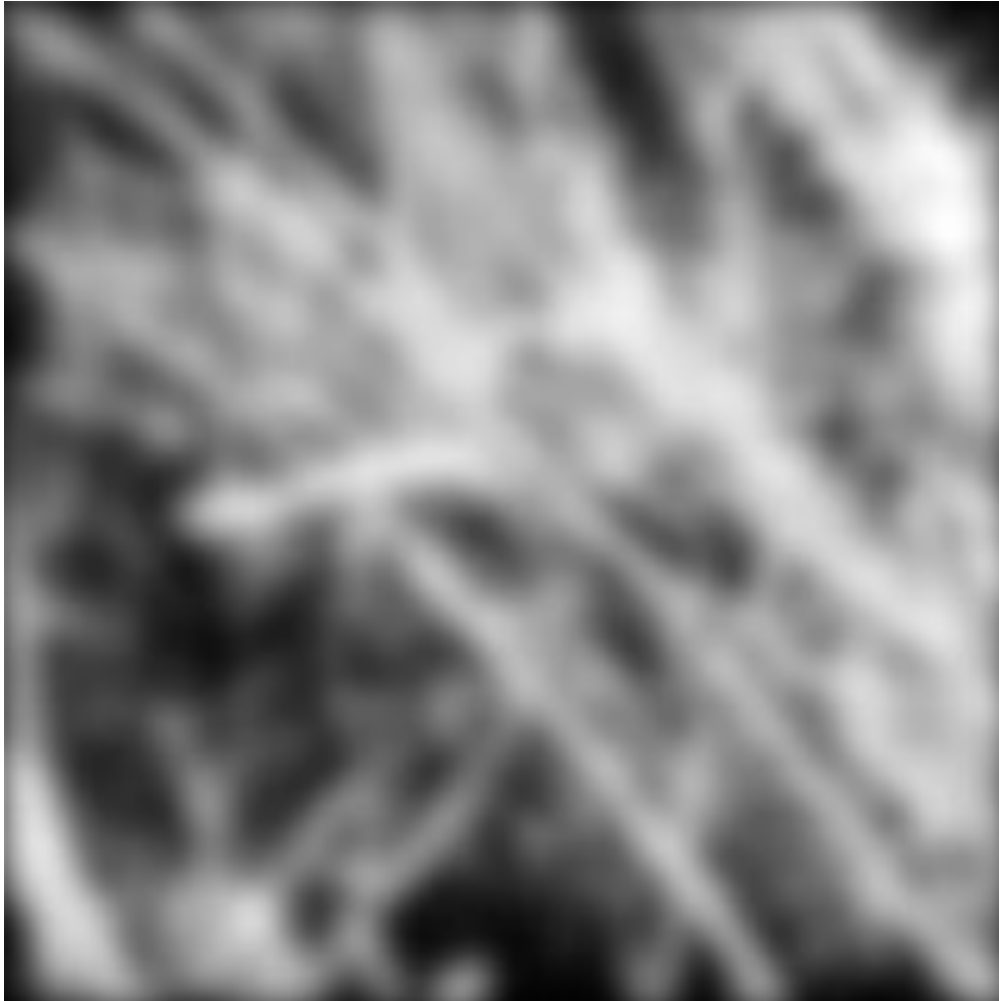


$|G(s_x, s_y)|$

Filtering

Why does the Gaussian give a nice smooth image, but the square filter give edgy artifacts?

Gaussian

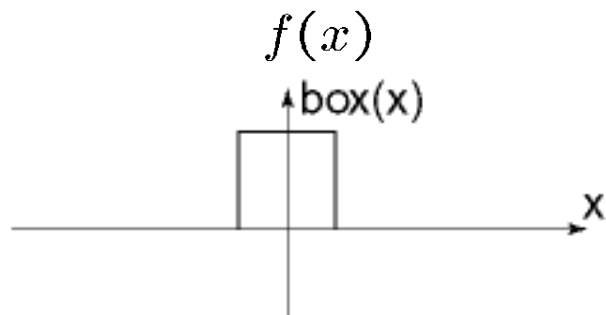


Box filter

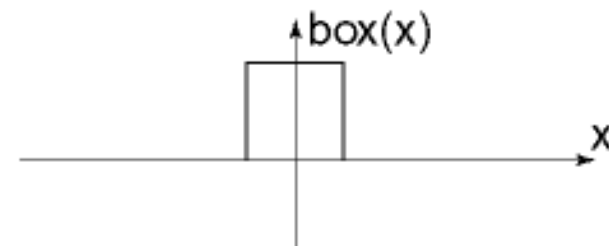
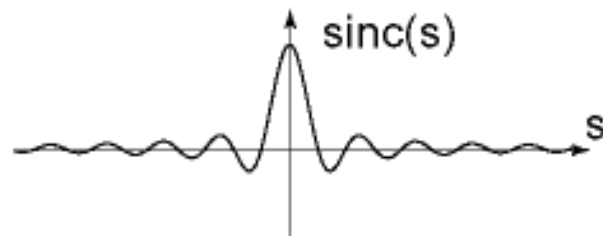
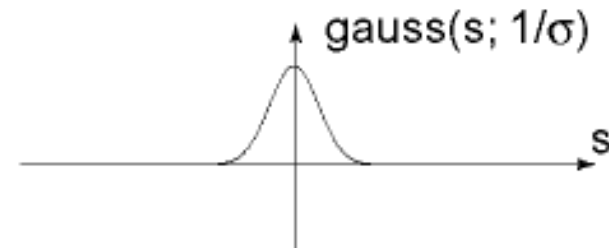
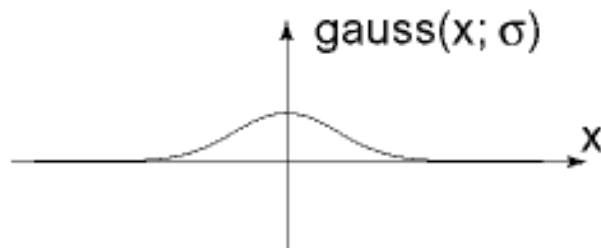
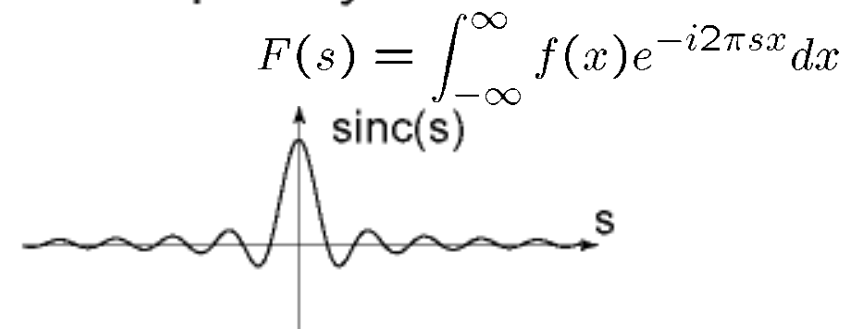


Fourier Transform pairs

Spatial domain



Frequency domain

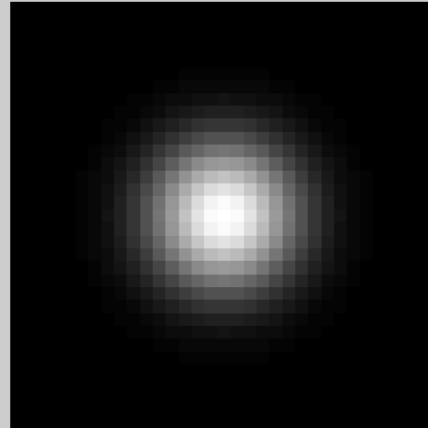


Gaussian

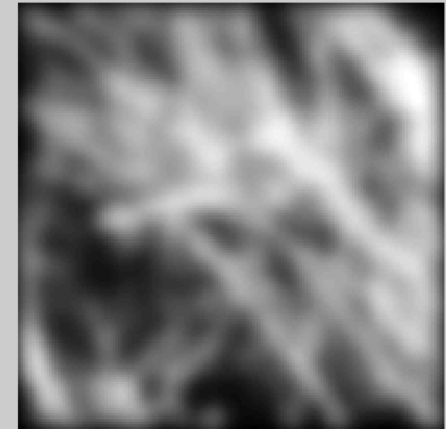
intensity image



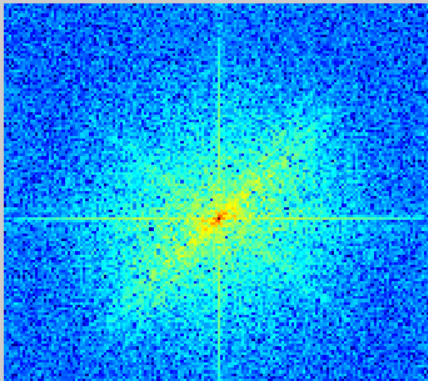
filter: gaussian



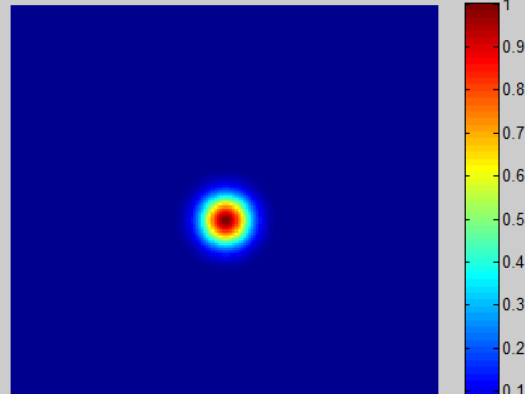
filtered image



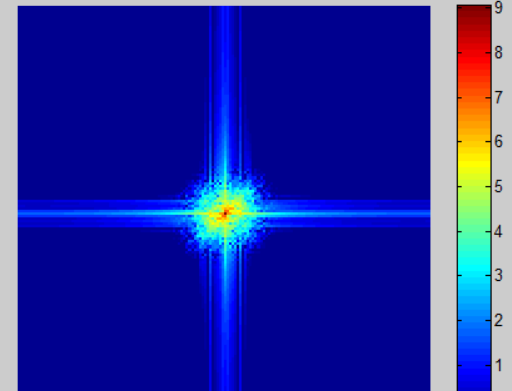
log fft magnitude of image



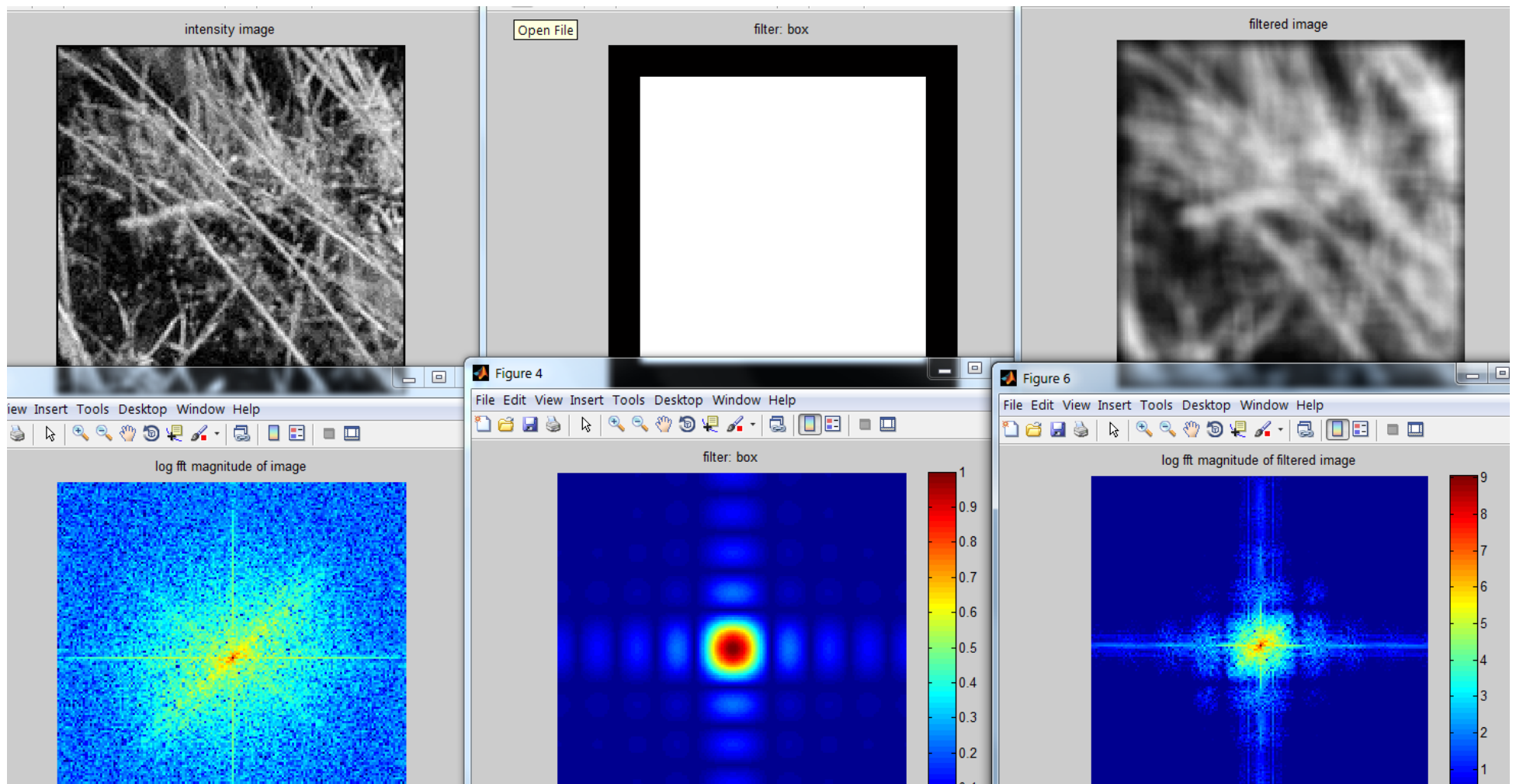
filter: gaussian



log fft magnitude of filtered image

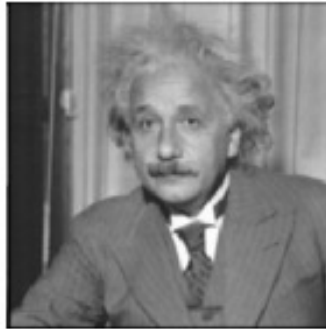
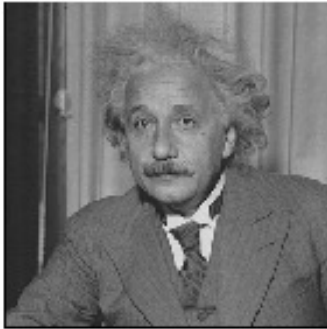


Box Filter

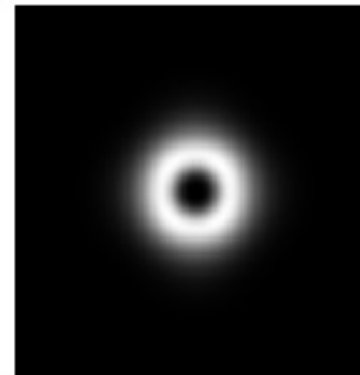
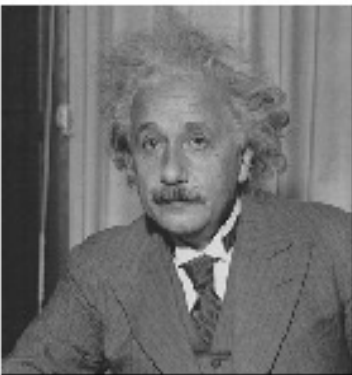


Low-pass, Band-pass, High-pass filters

low-pass:



High-pass / band-pass:



Low Pass vs. High Pass filtering

Image



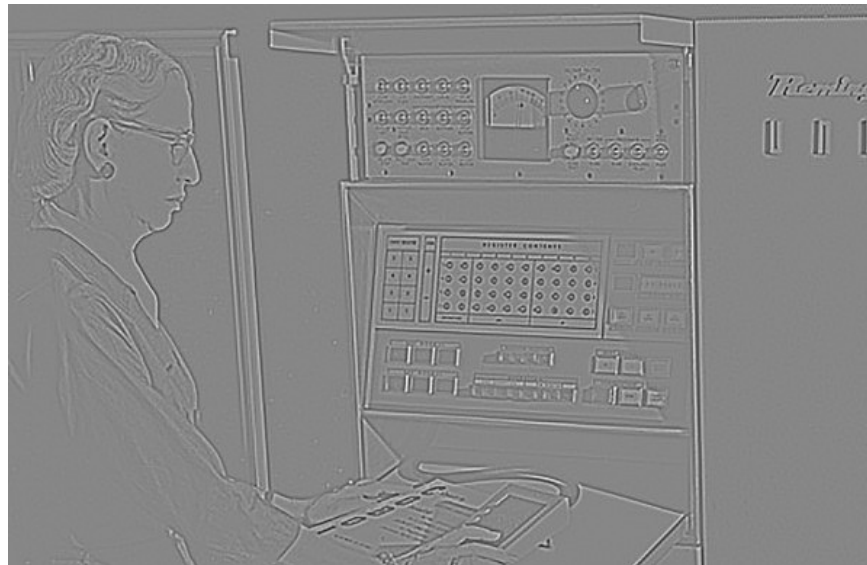
Smoothed



—

Details

=



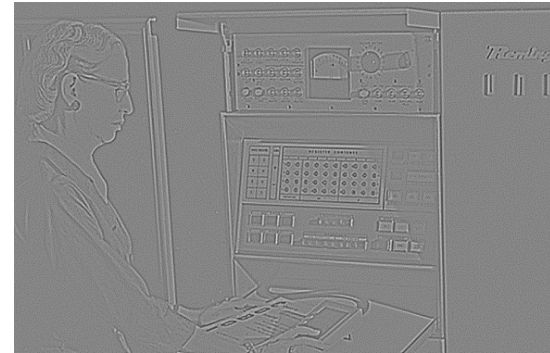
Filtering – Sharpening

Image



$+\alpha$

Details



“Sharpened” $\alpha=2$

$=$



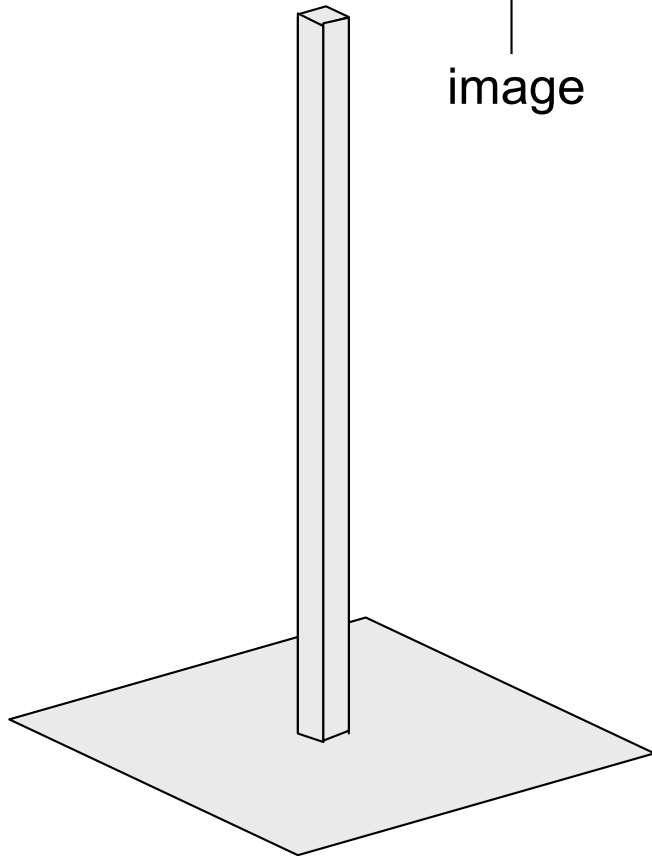
Unsharp mask filter (= sharpening filter)

$$f + \alpha(f - \underbrace{f * g}_{\text{blurred image}}) = f + \alpha f * (\underbrace{e - g}_{\text{unit impulse (identity)}})$$

↑
image

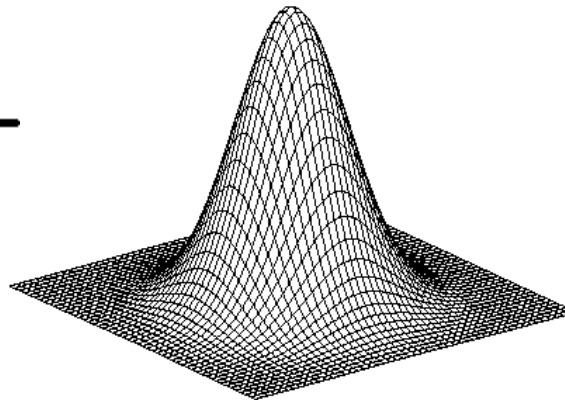
↑
blurred
image

↑
unit impulse
(identity)



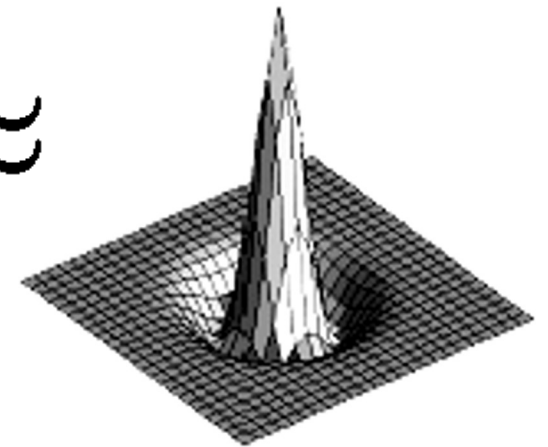
unit impulse

—



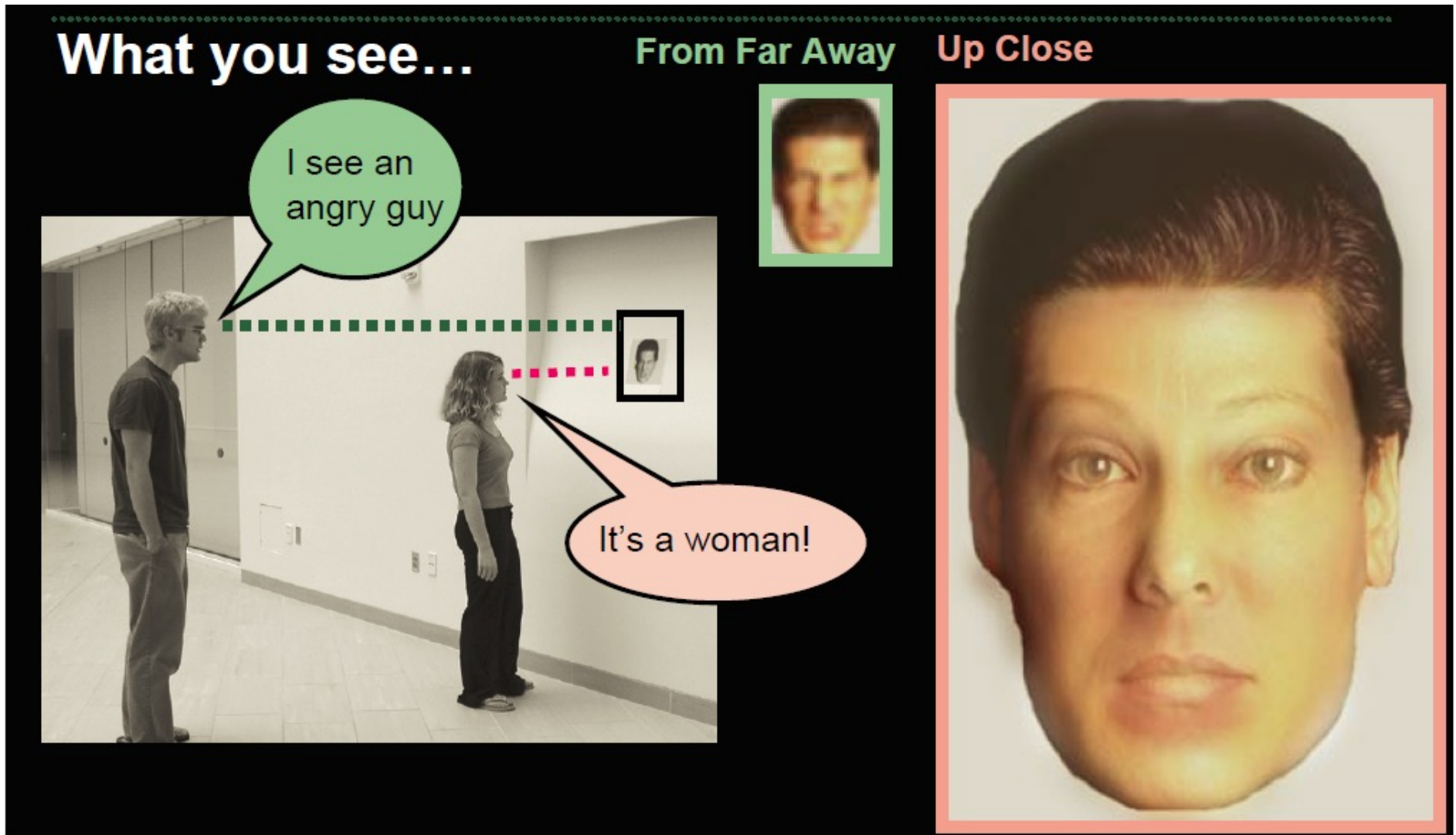
Gaussian

≈



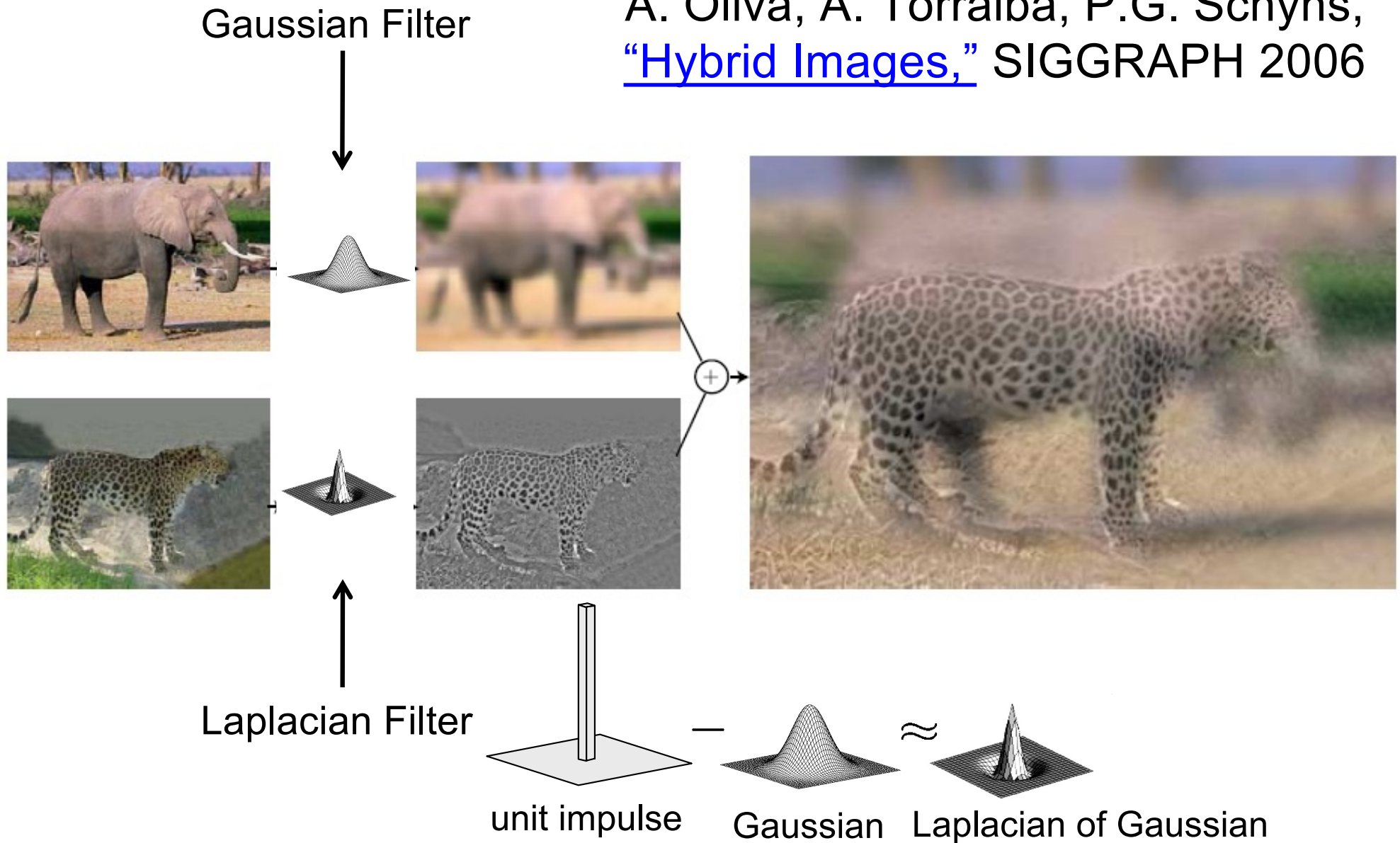
Laplacian of Gaussian

Application: Hybrid Images

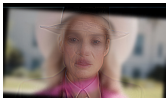


Application: Hybrid Images

A. Oliva, A. Torralba, P.G. Schyns,
[“Hybrid Images,”](#) SIGGRAPH 2006



Yesteryear's homework



Barbenheimer
CS180: Nicolas Rault-Wang

5 min recap

Fourier Transform in 5 minutes: The Case of the Splotched Van Gogh, Part 3

<https://www.youtube.com/watch?v=JciZYrh36LY>